Variational Methods Applied to Modelling

Lecture notes

ENSIMAG 2A MMIS and Master 1 AM

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Warning: the objective of this document is only to summarize some key notions and gather some lecture notes. It is thus far from being exhaustive nor self-sufficient.
Notice that the slides presented in class do not follow exactly the content of this document.

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Chapter 1

Introduction: from finite difference to finite element methods

In the first semester, you discussed finite difference methods for PDE solving. A major drawback of these methods is that they rely on a Cartesian grid, which makes it difficult to apply them to non-rectangular domains. These methods were directly based on a discretization of the equation via a domain discretization. Variational methods discretize the space of the unknown solutions rather than the domain on which the equation is stated (although in the finite element method we will see that to discretize the solution space, we rely on a discretization of the domain). This new method does not work directly on the equation but on an equivalent alternative formulation of the equation.

Consider, for instance, the boundary problem associated with the simplest second order partial differential equation,

$$\begin{cases} -\Delta u = f & \text{on } \Omega \\ u = 0 & \text{sur } \partial\Omega \end{cases} \tag{1.1}$$

where Ω is a bounded domain of $\mathbb{R}^N$, $N = 2$ or 3 in applications. We call classical solution of this problem a function $u : \Omega \to \mathbb{R}$ such that $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^1(\overline{\Omega})$ and verifying (1.1). Let $v \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^1(\overline{\Omega})$ vanishing on $\partial\Omega$, and multiply the first equation of (1.1) by v, then integrate by parts:

$$\int_\Omega -\Delta u \, v dx = \int_\Omega f v dx.$$

From the Green formula and homogeneous boundary conditions,

$$\int_\Omega \nabla u \cdot \nabla v dx = \int_\Omega f v dx,$$

for all function v belonging to $\mathcal{C}^2(\Omega) \cap \mathcal{C}^1(\overline{\Omega})$ and vanishing on the boundary.

Let us denote by V the space of such functions. The formulation of problem (1.1) in

$$\text{Find } u \in V, \ \forall v \in V, \quad \int_\Omega \nabla u \cdot \nabla v dx = \int_\Omega f v dx \tag{1.2}$$

will be refered as its variational formulation. This denomination comes from the fact that it amounts to writing the variation of a certain energy defined on a space of functions. From a

mechanical point of view, and it is this community that was first interested in this formulation, this corresponds to writing a principle of virtual work: the work of the force f during a displacement v in the right and side of (1.2) corresponds to a variation of the system energy in the left-hand side. Observe the duality between (1.1) which can be rewritten as

$$\text{Find } u \in V,\ \forall x \in \Omega, \quad -\Delta u(x) = f(x) \tag{1.3}$$

and the variational form (1.2), between the points space Ω and the functions space V.

The existence of solutions to (1.2) is obtained by the Lax-Milgram theorem, which gives sufficient conditions for an equation of the form

$$\text{Find } u \in V,\ \forall v \in V, \quad a(u,v) = \ell(v) \tag{1.4}$$

to have an unique solution. In that equation a is a bilinear form and ℓ a linear form. This theorem is proved for an Hilbert space V. To give a meaning to each term of (1.2), it suffices that u and v both belong to $\mathcal{C}^1(\Omega) \cap \mathcal{C}^0(\overline{\Omega})$. This is another advantage of the variational form: the *a priori* requested regularity to give a meaning to the equation is weaker than in the classical formulation.

The drawback is that the space $V = \left\{u \in \mathcal{C}^1(\Omega) \cap \mathcal{C}^0(\overline{\Omega}),\ u = 0 \text{ sur } \partial\Omega\right\}$ endowed with the natural scalar product is not an Hilbert (it is not a complete space) $\Rightarrow$ the completion of such spaces will give Sobolev spaces. In those extension of spaces of smooth functions, assignating a value to the restriction of a function to the boundary is a problem, so we will spend some time to describe how to cope with that.

After this introduction to Sobolev spaces and their main properties, we will introduce the Lax-Milgram theory and see how to fomulate several boundary problems in this framework. The approximation process is directly done on the space V. We will write an approximate variational formulation on a subspace of finite dimension of V :

$$\text{Find } u_h \in V_h, \forall v_h \in V_h, \quad a(u_h, v_h) = \ell(v_h). \tag{1.5}$$

This subspace will be expected to approach V in a precise meaning as $h \to 0$. On this subspace, the problem is reduced to the solution of a linear system whose unknowns are the components of the unknown in a basis of V_h. The central question of numerical analysis is the estimation of the error that is made by replacing a continuous problem (without an explicit solution in general) by an approximate problem that can be implemented on machine. The variational method offers a very elegant framework by turning the error estimate between the approximate solution $u_h \in V_h$ and the exact solution $u \in V$ to an approximation problem of u by functions of V_h. The different finite element methods which stem correspond to different choices of the space V_h and to the study of the corresponding approximation error. We will conclude this part of the lecture on these issues.

To sum up, the variational approach of which the finite elements are a particular case corresponds to a path followed for the rest of the lecture:

1. Choice of space V and variational formulation of the (initial) boundary value problem.
2. Application of Lax-Milgram theorem on V.
3. Discretization of V and approximate variational formulation.
4. Error estimate and convergence result.
5. Numerical implementation on computer (with FreeFem++).

Chapter 2

Brief introduction to Sobolev spaces

2.1 Definitions

Let $\Omega \subset \mathbb{R}^N$ be an open set. We denote by $H^1(\Omega)$ the distributions of $L^2(\Omega)$ with all their first order partial derivatives belonging to $L^2(\Omega)$. Thus denoting the partial derivatives (as distributions) $\partial_\alpha := \frac{\partial}{\partial x_\alpha}$

$$H^1(\Omega) = \left\{u \in L^2(\Omega),\ \partial_\alpha u \in L^2(\Omega),\ \alpha = 1 \dots N\right\}.$$

Proposition 2.1. $H^1(\Omega)$ is an Hilbert space for the scalar product

$$\langle u, v\rangle_1 = \int_\Omega uv + \nabla u \cdot \nabla v dx.$$

We will denote by $\|\cdot\|_1$ the associated norm.

Proof. Let (u_n) be a Cauchy sequence in H^1. Then (u_n) is a Cauchy sequence in L^2 which is complete. As well, $(\partial_\alpha u_n)$ is a Cauchy sequence in L^2 thus these sequences are convergent. Let u and u_α their respective limits. By taking $\phi \in \mathcal{D}(\Omega)$ we get $\langle \partial_\alpha u_n, \phi\rangle = -\langle u_n, \partial_\alpha \phi\rangle$. The first term of this equality goes to $\langle u_\alpha, \phi\rangle$ and the second to $-\langle u, \partial_\alpha \phi\rangle$. The two limits being equal we have $u_\alpha = \partial_\alpha u$. Thus $u \in H^1$ and $u_n \to u$ in H^1. $\square$

Alternative definition. For the whole space:

$$H^1(\mathbb{R}^N) = \left\{u \in \mathcal{S}'(\mathbb{R}^N), \xi \to (1+|\xi|^2)^{\frac{1}{2}}\hat{u}(\xi) \in L^2(\mathbb{R}^N)\right\}$$

This definition extends to Sobolev spaces of order s by setting

$$H^s(\mathbb{R}^N) = \left\{u \in \mathcal{S}'(\mathbb{R}^N), \xi \to (1+|\xi|^2)^{\frac{s}{2}}\hat{u}(\xi) \in L^2(\mathbb{R}^N)\right\}.$$

For an integer s entier, this is equivalent to require that all partial derivatives up to order s are in L^2.

Let us use this definition to solve a first PDE on $\mathbb{R}^N$, thus without boundary conditions. Let $f \in H^s(\mathbb{R}^N)$ and $k \neq 0$, we seek $u \in \mathcal{S}'(\mathbb{R}^N)$ solution of

$$-\Delta u + k^2 u = f.$$

By fourier transform we find

$$(|\xi|^2 + k^2)\hat{u} = \hat{f}$$

so that $\hat{u} = \frac{1}{k^2+|\xi|^2}\hat{f}(\xi)$. We thus have

$$(1+|\xi|^2)^{\frac{s+2}{2}}|\hat{u}(\xi)| \leq C(1+|\xi|^2)^{\frac{s}{2}}|\hat{f}| \in L^2(\mathbb{R}^N).$$

We therefore obtain a solution $u \in H^{s+2}(\mathbb{R}^N)$.

Problem. How to define the boundary value of a function which might not be continuous ? In dimension 1, functions from H^1 are continuous: if $u \in H^1(a,b)$ then $u' \in L^2(a,b)$. Let $v(x) = \int_a^x u'(s)ds$. The function v is (uniformly) continuous, since

$$|v(x) - v(y)| = \left|\int_y^x u'(s)ds\right| \leq |x-y|^{\frac{1}{2}}\|u'\|_{L^2}$$

For $\phi \in \mathcal{D}(]a,b[)$,

$$\int_a^b v(x)\phi'(x)dx = \int_{x=a}^b \int_a^x u'(y)\phi'(y)dydx = \int_{y=a}^b \int_{x=y}^b u'(y)\,\phi'(x)dxdy = -\int_a^b u'(y)\phi(y)dy.$$

Thus $v' = u'$ in the distribution sense, which means that v and u are equal up to u constant. As v is continuous, u too. This is wrong in higher dimension. A classical counter-example is given in $\mathbb{R}^2$ by $x \to (\ln|x|)^\alpha$ which is in H^1 of a unit ball centered in 0 for $0 < \alpha < \frac{1}{2}$ while non finite in 0.

Case of homogeneous conditions. We define the space

$$H_0^1(\Omega) = \overline{\mathcal{D}(\Omega)}^{H^1(\Omega)}$$

In other words, H_0^1 is the set of limits of sequence of $\mathcal{D}(\Omega)$ functions (for the H^1 norm).
Like those are of compact support, *if all goes well* this should give zero on the boundary in a generalized sense. If everything goes well, because we recall that

$$\overline{\mathcal{D}(\Omega)}^{L^2(\Omega)} = L^2$$

So *you can not define* the value of a function of L^2 at the boundary in general, because it can be approached as close as needed by a function which vanishes at the boundary. The results that follow will prove that this is not the case for the functions of H^1.
General solution The idea is to define the boundary value as the limit of restriction to the boundary of a sequence of smooth functions approaching the function in H^1. We will need to know precisely if there are such sequences of smooth approximating functions.

2.2 The H_0^1 space

We defined that space as the closure of $\mathcal{D}(\Omega)$ for the H^1 norm. Those functions verify a fundamental inequality for what follows.

Theorem 2.1 (Poincaré inequality)**.** Let Ω a bounded open subset of $\mathbb{R}^N$. The there exists $C(\Omega) > 0$ depending only of Ω such that

$$\forall u \in H_0^1(\Omega), \quad \|u\|_{L^2(\Omega)} \leq C(\Omega)\|\nabla u\|_{L^2(\Omega)}.$$

Proof. Since the domain is bounded, we can assume that it is between two hyperplanes. Executing a change of coordinates we assume that these hyperplanes are $x_N = \alpha$ and $x_N = \beta$. We write $x \in \mathbb{R}^N$ as $x = (x', x_N)$, and we consider a function $u \in \mathcal{D}(\Omega)$. Finally we denote $\widetilde{u}$ the extension of u to $\mathbb{R}^N$ by 0. So

$$\|u\|^2_{L^2(\Omega)} = \|\widetilde{u}\|^2_{L^2(\mathbb{R}^N)} = \int_\alpha^\beta \int_{\mathbb{R}^{N-1}} \widetilde{u}(x', x_N)dx'dx_N.$$

Since $\widetilde{u}$ is a smooth function of x_N we have

$$\widetilde{u}(x', x_N) = \widetilde{u}(x', \alpha) + \int_\alpha^{x_N} \frac{\partial \widetilde{u}}{\partial x_N}(x', s)ds \text{ with } \widetilde{u}(x', \alpha) = 0.$$

On the other hand,

$$\left|\int_\alpha^{x_N} \frac{\partial \widetilde{u}}{\partial x_N}(x', s)ds\right| \leq \left(\int_\alpha^{x_N} 1^2 ds\right)^{\frac{1}{2}} \left(\int_\alpha^{x_N} (\frac{\partial \widetilde{u}}{\partial x_N})^2(x', s)ds\right)^{\frac{1}{2}},$$

thus

$$\widetilde{u}(x', x_N)^2 \leq (x_N - \alpha) \int_\alpha^{x_N} (\frac{\partial \widetilde{u}}{\partial x_N})^2(x', s)ds$$

and therefore

$$\int_{\mathbb{R}^{N-1}} \widetilde{u}^2(x', x_N)dx \leq (x_N - \alpha) \int_{\mathbb{R}^N} (\frac{\partial \widetilde{u}}{\partial x_N})^2(x)dx.$$

Finally by integrating from α to β we get

$$\|\widetilde{u}\|^2_{L^2} \leq \frac{(\beta - \alpha)^2}{2} \|\frac{\partial \widetilde{u}}{\partial x_N}\|^2_{L^2}.$$

The result follows for functions of $\mathcal{D}(\Omega)$ and then by density in $H_0^1(\Omega)$. □

Remark 2.1.

- This result shows in particular that for Ω bounded, $H_0^1(\Omega)$ is a proper subspace of $H^1(\Omega)$. Indeed, the constant function equal to 1 belongs to $H^1(\Omega)$ and does not belong to $H_0^1(\Omega)$, otherwise it would fulfill the above equality which is absurd because its gradient is zero.
- The above theorem is valid as soon as Ω is bounded in one direction.
- If Ω is a bounded open set, the semi-norm $\|\nabla u\|_{L^2}$ is a norm on H_0^1 equivalent to the H^1 norm. For the associated scalar product (and therefore also for that of H^1), H_0^1 is an Hilbert space.

2.3 Notion of trace

As we saw, H^1 function need not to be continuous. To define their value on the boundary of an open set, we will approach them by smooth functions. The first result is that $\mathcal{D}(\mathbb{R}^N)$ is dense in $H^1(\mathbb{R}^N)$. The proof of this result is performed by truncation and regularization. We first show that the set of bounded functions of $H^1(\mathbb{R}^N)$ is dense in $H^1(\mathbb{R}^N)$, then we regularize those function by convolution by smooth functions. Then, if Ω is an open set whose boundary is piecewise $\mathcal{C}^1$ (meaning that locally its boundary is the graph of a $\mathcal{C}^1$ function), in particular if $\Omega = \mathbb{R}^N_+$ then $\mathcal{D}(\overline{\Omega})$ is dense in $H^1(\Omega)$. As it is trivial to assign to a smooth function its value on the boundary, we can define the boundary value of a H^1 function f as the limit of boundary values of a sequence of functions f_n of $\mathcal{D}(\overline{\Omega})$ which converges to f in H^1. For this to make sense, we have to prove that the mapping of a function to its restriction on the boundary is continuous for the H^1 norm (and not for the L^2 norm of course). More precisely, we prove the following theorem:

Theorem 2.2 (Trace theorem)**.** Let Ω be an open bounded subset of $\mathbb{R}^N$ with piecewise $\mathcal{C}^1$ boundary $\Gamma := \partial\Omega$. The the trace map

$$\gamma_0 : \begin{cases} \mathcal{D}(\overline{\Omega}) \to \mathcal{C}^0(\Gamma) \\ v \to \gamma_0(v) = v_{|\Gamma} \end{cases}$$

can be extended continuously into a continuous linear application of $H^1(\Omega)$ to $L^2(\Gamma)$. We thus have:

$$\exists C > 0, \quad \forall v \in H^1(\Omega), \quad \|\gamma_0(v)\|_{L^2(\Gamma)} \leq C\|v\|_{H^1(\Omega)}.$$

Remark 2.2. We will see that the kernel of γ_0 is actually H^1_0. The image of γ_0 is not $L^2(\Gamma)$, but an intermediate space between $L^2(\Gamma)$ and $H^1(\Gamma)$. We have defined Sobolev spaces of non-integer exponents only for the whole space, but this intermediate space would correspond to $H^{\frac{1}{2}}(\Gamma)$, defined by locally straightening Γ.

Proof. To prove the trace theorem, we restrict ourselves to the case $\Omega = \mathbb{R}^N_+$, for which we want to define a trace on the hyperplane $X_N = 0$. The case of a generic domain Ω is addressed using local maps: we consider a part of $\partial\Omega$ and we straighten it thanks to a diffeomorphism into a part of $\mathbb{R}^N_+$ on which we apply the above result. Then the parts are glued together. When $\Omega = \mathbb{R}^N_+$, we will show that for $v \in \mathcal{D}(\overline{\mathbb{R}^N_+})$,

$$\|v(\cdot, 0)\|_{L^2(\mathbb{R}^{N-1})} \leq \|v\|_{H^1(\mathbb{R}^N_+)}.$$

Indeed if $v \in \mathcal{D}(\overline{\mathbb{R}^N_+})$, we have

$$v(x', 0)^2 - 0 = \int_{+\infty}^{0} \frac{\partial}{\partial x_N} v(x', x_N)^2 dx_N = -2\int_0^{+\infty} v(x', x_N)\frac{\partial v}{\partial x_N}(x', x_N)dx_N.$$

Thus

$$v(x', 0)^2 \leq 2\left(\int_0^{+\infty} v(x', x_N)^2 dx_N\right)^{\frac{1}{2}} \left(\int_0^{+\infty} \frac{\partial v}{\partial x_N}(x', x_N)^2 dx_N\right)^{\frac{1}{2}}$$
$$\leq \int_0^{+\infty} v(x', x_N)^2 dx_N + \int_0^{+\infty} \frac{\partial v}{\partial x_N}(x', x_N)^2 dx_N$$

which upon integration on R_+^{N-1} gives the announced inequality. Since $\mathcal{D}(\overline{\mathbb{R}_+^N})$ is dense in $H^1(\mathbb{R}_+^N)$, this inequality extends to functions of this space and one obtains, for these functions, a trace in L^2 of the boundary. Note that this is not the case for the functions of $L^2(\mathbb{R}_+^N)$: for these we can not define a value on the boundary. □

Corollary 2.1. $H_0^1(\Omega) = \left\{v \in H^1(\Omega), \quad \gamma_0(v) = 0_{L^2(\Gamma)}\right\}$.

Proof. Indeed, since the functions of H_0^1 are the functions limits of $\mathcal{D}(\Omega)$ for the topology of H^1 and they are cancelling on $\partial\Omega$, the continuitty of γ_0 implies that the limit function has a zero trace. Inclusion in one way is therefore proven. The other way is admitted. □

2.4 Green formula in Sobolev Spaces

The following result implies the Green formula.

Theorem 2.3 (Integration by parts in dimension N)**.** Let Ω be an open bounded subset of $\mathbb{R}^N$ with piecewise $\mathcal{C}^1$ boundary. Then if u and v are $H^1(\Omega)$ function, it holds

$$\int_\Omega (\partial_{x_i} u) v dx = -\int_\Omega u(\partial_{x_i} v) dx + \int_\Gamma u v n_i d\sigma, \qquad i = 1, \ldots, N$$

where n_i is i-th component of the outward unit normal to Ω.

Proof.
The formula is true for the smooth functions: Exercise! In $\mathbb{R}^2$, first consider as Ω the triangle $\{(x, y),\ x > 0, y > 0, x + y < 1\}$. Then on a triangle image of the former by diffeomorphism. Then by the theorem below (for triangles) by collecting these triangles, over any polygonal domain. It's already good! Finally, the continuity of trace map allows to pass to the limit on two sequences of smooth functions approaching the given functions. □

Application. Assume that we have a discretization of Ω in subsets Ω_i, $i = 1, \ldots, M$ of piecewise $\mathcal{C}^2$ boundary, so that $\overline{\Omega} = \bigcup_{i=1}^M \overline{\Omega_r}$ and $\Omega_i \cap \Omega_j = \emptyset$ for $i \neq j$. Then:

Proposition 2.2. Let v be a continuous function on $\overline{\Omega}$ such that its restriction to Ω_i is in $H^1(\Omega_i)$, for all $i = 1, \ldots, M$. Then $v \in H^1(\Omega)$.

Proof. Let $w_i \in L^2(\Omega)$ such that its restriction to Ω_j is equal to $\partial_{x_i}(v_{|\Omega_j})$, which is L^2 by assumption. Then take $\phi \in \mathcal{D}(\Omega)$, from the above theorem we have

$$\int_\Omega w_i \phi dx = \sum_{j=1}^M \int_{\Omega_j} \partial_{x_i}(v_{|\Omega_j}) \phi dx = -\sum_{j=1}^M \int_{\Omega_j} v_{|\Omega_j} \partial_{x_i} \phi dx + \sum_{j=1}^M \int_{\partial\Omega_j} v \phi n_i^j dx,$$

wheren_i^j is i-th component of the outward unit normal to Ω. Since v is continuous, the last sum vanishes because, for two triangles with one side in common, the two outgoing normals are opposite. We have therefore in the sense of distributions

$$\langle w_i, \phi \rangle = -\langle v, \partial_{x_i} \phi \rangle = \langle \partial_{x_i} v, \phi \rangle .$$

As $w_i \in L^2(\Omega)$ we deduce that $\partial_{x_i} v \in L^2(\Omega)$ for each i an thus that $v \in H^1(\Omega)$. □

Definition 2.1 (Sobolev spaces of order m).
Let
$$H^m(\Omega) = \left\{ v \in L^2(\Omega), \quad \partial^\alpha v \in L^2(\Omega), \quad |\alpha| \leq m \right\},$$
where
$$\partial^\alpha v = \frac{\partial^{|\alpha|} v}{\partial_{x_1}^{\alpha_1} \cdots \partial_{x_N}^{\alpha_N}}, \qquad |\alpha| = \alpha_1 + \cdots + \alpha_N.$$
It is an Hilbert space for
$$(u, v)_m = \int_\Omega \sum_{|\alpha| \leq m} \partial^\alpha u \partial^\alpha v dx,$$
and the associated norm is denoted by $\| \cdot \|_m$.

If Ω is an open bounded set of piecewise $\mathcal{C}^1$ boundary Γ, as $H^2(\Omega) \subset H^1(\Omega)$ we can define the trace of an element of H^2 on the boundary as before. But as $\partial_{x_i} v \in H^1(\Omega)$ for $v \in H^2(\Omega)$ we may also define the trace of
$$(\partial_n v)_{|\Gamma} = \sum_{i=1}^N \gamma_0(\partial_{x_i} v) \in L^2(\Gamma)$$
which is involved in the variational formulation of many boundary problems.

Theorem 2.4 (Trace theorem for H^2). Let Ω be an open bounded subset of $\mathbb{R}^N$ with piecewise $\mathcal{C}^1$ boundary $\partial\Omega$. Then the trace map defined from $\mathcal{D}(\overline{\Omega})$ to $\mathcal{C}^0(\Gamma) \times \mathcal{C}^0(\Gamma)$ by
$$v \to (\gamma_0 v, \gamma_1 v) = \left(v_{|\Gamma}, (\partial_n v)_{|\Gamma} \right)$$
extends into a continuous linear map from $H^2(\Omega)$ to $L^2(\Gamma) \times L^2(\Gamma)$ (in fact with range $H^{\frac{3}{2}}(\Gamma) \times H^{\frac{1}{2}}(\Gamma)$, restriction to the boundary removes $\frac{1}{2}$). In particular, this implies
$$\exists C > 0, \quad \forall v \in H^2(\Omega), \quad \|\gamma_0(\partial_n v)\|_{L^2(\Gamma)} \leq C \|v\|_{H^2(\Omega)}.$$

Note there no piecewise $\mathcal{C}^2$ regularity required on the boundary; indeed we obtain this result by applying the trace theorem in H^1 for the partial derivatives of the H^2 functions which needs only piecewise $\mathcal{C}^1$ regularity. With this notion of trace of H^2 function on the boundary, we can prove the following Green formula for Sobolev functions.

Theorem 2.5 (Green formula in H^2). Let Ω be an open bounded subset of $\mathbb{R}^N$ with piecewise $\mathcal{C}^1$ boundary, $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$. Then
$$\int_\Omega (\Delta u) v dx = -\int_\Omega \nabla u \cdot \nabla v dx + \int_\Gamma (\partial_n u) v d\sigma.$$

Proof. We apply the integration by parts formula in dimension N with $\partial_{x_i} u$ in place of u and we sum on i. □

To conclude this short introduction to Sobolev spaces, we will give a regularity result which says that when m is large enough, depending on the dimension, the functions of H^m become continuous:

Theorem 2.6 (Regularity of H^m functions)**.** If Ω is an open bounded subset of $\mathbb{R}^N$ of piecewise $\mathcal{C}^1$ boundary, then for $m > \frac{N}{2}$, we have $H^m(\Omega) \subset \mathcal{C}^0(\overline{\Omega})$ with continuous injection, which means

$$\exists C > 0,\ \forall u \in H^m(\Omega),\ \sup_{x\in\Omega} |u(x)| \leq C\|u\|_m.$$

Proof. A fairly technical result (reflection method, see [4] page 272 or [5] page 158) shows under the hypotheses of the theorem that there exists a continuous linear extension operator from $H^m(\Omega)$ to $H^m(\mathbb{R}^N)$. Let $u \in H^m(\Omega)$, we denote by U its extension. So there exists a constant C independent of u such that

$$\|U\|_{H^m(\mathbb{R}^N)} \leq C\|u\|_{H^m(\Omega)}.$$

Then write the Fourier transform of U in the form

$$\widehat{U}(\xi) = (1+|\xi|^2)^{-\frac{m}{2}}(1+|\xi|^2)^{\frac{m}{2}}\widehat{U}(\xi)$$

which makes appearing $\xi \to (1+|\xi|^2)^{\frac{m}{2}}\widehat{U}(\xi)$ which is in L^2 according to the characterization of H^s by Fourier transform (p. 3), and $\xi \to (1+|\xi|^2)^{-\frac{m}{2}}$ which is in L^2 if and only if $m > \frac{N}{2}$ (exercise: change to "polar" coordinates in $\mathbb{R}^N$). So in this case $\widehat{U} \in L^1$ and according to the properties of the Fourier transform, $U \in \mathcal{C}^0(\mathbb{R}^N)$. Since

$$U(x) = \frac{1}{(2\pi)^{N/2}} \int_{\mathbb{R}^N} e^{ix\cdot\xi}\widehat{U}(\xi)d\xi,$$

we have $|U(x)| \leq C\|\widehat{U}\|_{L^1(\mathbb{R}^N)} \leq C\|U\|_{H^m(\mathbb{R}^N)}$. As U is an extension, we have

$$\sup_{x\in\Omega} |u(x)| \leq \sup_{x\in\mathbb{R}^N} |U(x)|$$

and this prolongation verifies

$$\|U\|_{H^m(\mathbb{R}^N)} \leq C\|u\|_{H^m(\Omega)}.$$

Hence the result. □

Finally, the following compactness result will be useful to us. A result of compactness of one space in another is often used in PDE theory when information is available on the norm of the elements of a sequence of elements of the space. If this one is injected in a compact way into a larger space, then this bounded sequence will contain a convergent subsequence for the norm of the large space, which makes it possible to pass to the limit in some cases (for a first application, see the proof of the Poincaré inequality in the case of functions H^1 vanishing only on some part of the boundary).

Theorem 2.7 (Rellich)**.** Let Ω be an open bounded open set of $\mathbb{R}^N$ of piecewise $\mathcal{C}^1$ boundary. Then $H^1(\Omega)$ injects compactly into $L^2(\Omega)$. That is to say that from any bounded sequence in H^1 we can extract a subsequence converging in norm in L^2. We have the same result for H^1_0, without regularity assumption on $\partial\Omega$.

Chapter 3

Elliptic boundary value problems

3.1 Variational formulation

3.1.1 Dirichlet problem for Laplace operator

Let Ω be an open bounded subset of $\mathbb{R}^N$ with piecewise $\mathcal{C}^1$ boundary and $f \in L^2(\Omega)$. We aim to solve the following problem: Find u such that

$$(D) \qquad \begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Suppose there exists a solution $u \in H^2(\Omega)$ to (D). Let $v \in H_0^1(\Omega)$, we obtain by multiplying the first equation of (D) by v and by integrating on Ω:

$$\int_\Omega -(\Delta u) v dx = \int_\Omega f v dx$$

so that from the Green formula,

$$\int_\Omega \nabla u \cdot \nabla v dx - \int_{\partial\Omega} \partial_n u v d\sigma = \int_\Omega f v dx$$

As $v = 0$ on the boundary we obtain what is called the variational formulation of (D):

$$(FVD) \qquad \text{Find } u \in H_0^1(\Omega), \forall v \in H_0^1(\Omega), \quad \int_\Omega \nabla u \cdot \nabla v dx = \int_\Omega f v dx.$$

Conversely, if $u \in H_0^1(\Omega) \cap H^2(\Omega)$ is a solution of (FVD) then the inverse reasoning shows that u is a solution of (D). The variational formulation is therefore equivalent for the regular solutions to the classical formulation of the Dirichlet problem. However, we see that it involves only first derivatives of the unknown. It therefore potentially allows to find generalized solutions of (D) when the classical solutions do not exist.

3.1.2 Neumann problem for Laplace operator

Still with the same assumptions on Ω, we now look for a function u such that

$$(N) \qquad \begin{cases} -\Delta u + u = f & \text{in } \Omega, \\ \partial_n u = 0 & \text{on } \partial\Omega. \end{cases}$$

Suppose there exists a solution $u \in H^2(\Omega)$ to (N). Let $v \in H^1(\Omega)$ be obtained by multiplying the first equation of (N) by v and by integrating on Ω:

$$\int_\Omega -(\Delta u)v + uvdx = \int_\Omega fvdx$$

so that from the Green formula,

$$\int_\Omega \nabla u \cdot \nabla v + uvdx - \int_{\partial\Omega} \partial_n uvd\sigma = \int_\Omega fvdx.$$

As $\partial_n u = 0$ on the boundary we obtain what is called the variational formulation of (N) :

$$(FVN) \qquad \text{Find } u \in H^1(\Omega),\ \forall v \in H^1(\Omega), \quad \int_\Omega \nabla u \cdot \nabla v + uvdx = \int_\Omega fvdx.$$

Conversely, if $u \in H^2(\Omega)$ is a solution of (FVN) then in particular if we choose $v \in \mathcal{D}(\Omega)$ in the formulation, $\int_\Omega \nabla u \cdot \nabla v + uvdx = \int_\Omega fvdx$ so that by using the Green formula, the boundary term vanishes because v has compact support,

$$\int_\Omega (-\Delta u)v + uvdx = \int_\Omega fvdx.$$

We thus have $-\Delta u + u = f$ in the distribution sense, and as f and u are in L^2, this equality takes place in this space and almost everywhere. Let us now take a test function v in $H^1(\Omega)$ and integrate the variational formulation (FVN) by parts. We get

$$\int_\Omega (-\Delta u)v + uvdx - \int_{\partial\Omega} \partial_n uvd\sigma = \int_\Omega fvdx,$$

so that by the L^2 equality obtained, $\int_{\partial\Omega} \partial_n uvd\sigma = 0$ for all $v \in H^1(\Omega)$. Hence the seeked boundary condition.

Remark 3.1. The difference between the Dirichlet problem and the Neumann problem is therefore in the treatment of the limit condition. In one case it is taken into account in the space definition, in the other hidden in the formulation. We will return to this difference after having examined the generic tool to solve problems posed in variational form.

3.2 Lax-Milgram theorem

The Lax-Milgram theorem (Peter Lax, American mathematician, born Budapest in 1926 and still alive in Feb. 2023, and Arthur Milgram, American mathematician 1912–1961) can be related to the Riesz representation theorem but extends to the case of non symmetric coercive bilinear forms.

Theorem 3.1. Let V be an Hilbert space and $a : V \times V \to \mathbb{R}$ a bilinear continuous and coercive form on $V \times V$ and $\ell : V \to \mathbb{R}$ a linear continuous form V, that is

(i) $\exists M > 0,\ \forall (u,v) \in V^2,\ |a(u,v)| \leq M\|u\|\|v\|$ (Continuity of a)

(ii) $\exists \alpha > 0,\ \forall u \in V\ a(u,u) \geq \alpha\|u\|^2$ (Coerciveness of a)

(iii) $\exists C > 0,\ \forall v \in V,\ |\ell(v)| \leq C\|v\|$ (Continuity of ℓ)

Then the problem: Find $u \in V$ such that $\forall v \in V,\ a(u,v) = \ell(v)$ has a unique solution.

Proof. Let $A : V \to V$ be the linear operator defined by $\langle Au, v\rangle = a(u,v)$ for all v in V. The existence of Au results from the Riesz theorem, since $v \to a(u,v)$ is linear continuous on V. The operator A is linear continuous because

$$\|Au\| = \sup_{v \in V\setminus\{0\}} \frac{|\langle Au, v\rangle|}{\|v\|} \leq \frac{M\|u\|\|v\|}{\|v\|} = M\|u\|.$$

On the other hand, according to the Riesz representation theorem, there exists $b \in V$ such that $\ell(v) = \langle b, v\rangle$, $\forall v \in V$. The problem is therefore to show that there exists $u \in V$ such that $Au = b$. We will proceed in an iterative way starting from any $u_0 \in V$ and defining, for $\rho > 0$, the inductive sequence

$$u_{k+1} = u_k - \rho(Au_k - b) = F(u_k).$$

Let us show that for $\rho > 0$ small enough, F is a strict contraction. We have

$$F(u) - F(v) = u - v - \rho(A(u - v))$$

thus

$$\begin{aligned}\|F(u) - F(v)\|^2 &= \|u - v\|^2 + \rho^2\|A(u - v)\|^2 - 2\rho\,\langle A(u - v), u - v\rangle \\ &= \|u - v\|^2 + \rho^2\|A(u - v)\|^2 - 2\rho a(u - v, u - v) \leq \|u - v\|^2 + M^2\rho^2\|(u - v)\|^2 - 2\rho\alpha\|u - v\|^2 \\ &\leq (1 - 2\rho\alpha + \rho^2M^2)\|u - v\|^2.\end{aligned}$$

Let us choose ρ such that $\rho < 2\frac{\alpha}{M^2}$. Then F is a strict contraction and from the fixed point theorem (valid on a complete space) the iterative scheme is convergent, which proves the theorem. □

3.3 Application to the former problems

3.3.1 Dirichlet problem

Recall that it amounts to find $u \in H_0^1(\Omega)$ such that for all $v \in H_0^1(\Omega)$,

$$\int_\Omega \nabla u \nabla v dx = \int_\Omega f v dx.$$

We therefore apply the above theorem with:

$$a(u,v) = \int_\Omega \nabla u \nabla v dx, \qquad \ell(v) = \int_\Omega f v dx,$$

on the space $V = H_0^1(\Omega)$ endowed with the norm $\|u\|_V = \|\nabla u\|_{L^2(\Omega)}$. Then a is continuous from by the Cauchy-Schwarz inequality

$$|a(u,v)| \leq \|\nabla u\| \|\nabla v\| = \|u\|_V \|v\|_V.$$

The bilinear form a is coercive because $a(u,u) = \|u\|_V^2$. And finally ℓ is continuous because

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq C(\Omega) \|f\|_{L^2(\Omega)} \|v\|_V$$

From the Poincaré inequality. The Lax-Milgram theorem thus ensures us the existence and uniqueness of the solution of the Dirichlet problem for the Laplace operator.

Remark 3.2. We could also choose the same space V but normed with the norm of H^1. Then the continuity of a and ℓ are trivial, and the inequality of Poincaré is used in the coerciveness of a.

3.3.2 Neumann problem

We leave the Neumann problem to the reader, the demonstration being even simpler than the former. In the form we have chosen, the bilinear form a is exactly the scalar product of H^1. We will rather deal with a more general case.

3.3.3 Mixed boundary value problem for an elliptic second order operator

Let Ω be a bounded open set with boundary Γ piecewise of class $\mathcal{C}^1$, Γ_0 and Γ_1 a partition of Γ: $\Gamma_0 \cup \Gamma_1$ and $\Gamma_0 \cap \Gamma_1 = \emptyset$. We then consider $V = \{u \in H^1(\Omega),\ \gamma_0 u = 0 \text{ on } \Gamma_0\}$,

$$a(u,v) = \int_\Omega \sum_{i,j=1}^N a_{ij} \partial_{x_j} u \partial_{x_i} v + a_0 u v dx$$

where a_{ij} and a_0 are $L^\infty(\Omega)$ functions, and the linear form

$$\ell(v) = \int_\Omega f v dx$$

is such that $f \in L^2(\Omega)$. The variational formulation under study is:

Find $u \in V$ such that $a(u,v) = \ell(v)$ for all $v \in V$

We suppose that there exists $\alpha > 0$ such that for all $\xi \in \mathbb{R}^N$, we have

$$\sum_{i,j} a_{ij} \xi_i \xi_J \geq \alpha |\xi|^2,$$

and $\alpha_0 > 0$ such as $a_0 \geq \alpha_0$ a.e. in Ω. The space V is an Hilbert, because it is a closed subspace of an Hilbert (the kernel of a continuous linear form, by the trace theorem) . The scalar product is that of H^1.

We can easily check the assumptions of Lax-Milgram theorem: a is continuous because

$$|a(u,v)| \leq N^2 \max_{i,j} \|a_{ij}\|_\infty \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + \|a_0\|_\infty \|u\|_{L^2} \|v\|_{L^2}$$
$$\leq \max(N^2 \max_{i,j} \|a_{ij}\|_\infty, \|a_0\|_\infty) \|u\|_{H^1} \|v\|_{H^1},$$

It is coercive because according to the assumptions made on the coefficients a_{ij} we have

$$a(u,u) \geq \alpha \|\nabla u\|_{L^2}^2 + \alpha_0 \|u\|_{L^2}^2 \geq \min(\alpha, \alpha_0) \|u\|_{H^1}.$$

Finally ℓ is continuous as in the Dirichlet problem before.

Therefore there exists a unique $u \in V$ solution of $a(u,v) = \ell(v)$ for all $v \in V$. But what problem have we solved? Suppose that the solution we have found is H^2 and verify what problem it correspond to in the classical sense. Since the variational formulation holds for all $v \in V$, this is in particular the case for any $v \in \mathcal{D}(\Omega)$. As in this case the value of v at the boundary is 0, the Green formula applied to the variational formulation

$$\int_\Omega \sum_{i,j} a_{ij} \partial_{x_i} u \partial_{x_j} v + a_0 u v dx = \int_\Omega f v dx$$

gives

$$\int_\Omega [-\sum_{i,j} \partial_{x_i}(a_{ij} \partial_{x_j} u) + a_0 u] v dx = \int_\Omega f v dx$$

which means in the distributions sense on Ω

$$-\sum_{i,j} \partial_{x_i}(a_{ij} \partial_{x_j} u) + a_0 u = f.$$

Since we have assumed the solution H^2 this equality actually takes place in L^2 and a.e. from the regularity of the coefficients a. What's going on at the boundary? We already know that $u = 0$ on Γ_0, because $u \in V$. But on Γ_1? To find out, let us use $v \in V$ in the variational formulation and apply Green again; This time the boundary terms disappear on Γ_0 where v vanishes but not on Γ_1:

$$\int_\Omega [-\sum_{i,j} \partial_{x_i}(a_{ij} \partial_{x_j} u) + a_0 u] v dx + \int_{\Gamma_1} (\sum_{i,j} a_{ij} \partial_{x_j} u) v n_i d\sigma = \int_\Omega f v dx.$$

But we just showed that in Ω, in the L^2 sense and a.e. we have $-\sum_{i,j} \partial_{x_i}(a_{ij} \partial_{x_j} u) + a_0 u = f$. So the terms in Ω cancel and we find

$$\int_{\Gamma_1} \sum_{i,j} a_{ij} (\partial_{x_j} u) n_i v d\sigma = 0$$

for all v in V. If we denote by $\partial_{n_A} u = \sum_{i,j} a_{ij} (\partial_{x_j} u) n_i$, we get as solved problem written in the classic form:

$$-\sum_{i,j} \partial_{x_i}(a_{ij} \partial_{x_j} u) + a_0 u = f \text{ dans } \Omega, \quad u = 0 \text{ sur } \Gamma_0, \quad \partial_{n_A} u = 0 \text{ sur } \Gamma_1.$$

The boundary condition on Γ_1 is only formal in general if u does not have the H^2 regularity.

Remark 3.3. 1. If we want to solve a non-homogeneous problem on Γ_1, for instance $\partial_n u = g$, we can simply add to ℓ the corresponding term: $\ell(v) = \int_\Omega f v dx + \int_{\Gamma_1} g v d\sigma$, for $g \in L^2(\Gamma_1)$. The continuity of ℓ comes from the trace theorem.

2. To solve the same problem but with $u = u_0$ on Γ_0, the simplest is to consider this boundary value as the trace of a function (still denoted by) $u_0 \in H^1(\Omega)$ and performing the unknown function change: $U = u - u_0$. This is equivalent to still consider the same space $H_0^1(\Omega)$ and $\ell(v) = \int_\Omega f v dx - a(u_0, v)$, and then solve $a(U, v) = \ell(v)$.

3. The Neuman-type conditions, as we have seen, are "hidden" in the formulation while those of Dirichlet are in space. Why not incorporate Neumann's conditions into space? Since they are of order 1, this could not be a space based on H^1, but rather on H^2 because then the normal derivative has a trace on the boundary. But then the bilinear form a would not be coercive for the H^2 norm.

4. A Poincaré inequality holds on V as soon as $\operatorname{mes}\Gamma_0 > 0$ and Ω is connex. In this case one can then take $\alpha_0 = 0$. To prove the inequality in this case, ie to show that there exists $C > 0$ such that
$$\forall u \in V,\ C\|u\|_{L^2} \leq \|\nabla u\|_{L^2},$$
let us note first that according to the definition of the norm H^1 it is equivalent to show that
$$\exists C > 0,\ \forall u \in V,\ C\|u\|_{H^1} \leq \|\nabla u\|_{L^2}.$$
Let us proceed by the absurd: suppose, on the contrary, for $C = \frac{1}{m}$ with $m \in \mathbb{N}^*$ that
$$\forall m \in N^*,\ \exists u_m,\ \frac{1}{m}\|u_m\|_{H^1} > \|\nabla u_m\|_{L^2}.$$
Let $v_m = \frac{u_m}{\|u_m\|_{H^1}}$, we have $\|v_m\|_{H^1} = 1$, and $\frac{1}{m} > \|\nabla v_m\|_{L^2}$ for all m. Hence, the sequence v_m remains in a bounded subset of H^1 (the unit ball!). By compactness of the injection of H^1 into L^2 this implies that there exists a subsequence $v_{\sigma(m)}$ converging in L^2: $V_{\sigma(m)} \to v$. On the other hand, we have $\nabla v_{\sigma(m)} \to 0$ in L^2 by the above bound. It is easy to show that $\nabla v_{\sigma(m)}$ converges to ∇v in the sense of distributions, and thus by uniqueness of the limit we have $\nabla v = 0$. Thus v is constant on each connected component of Ω. Since, on the other hand, v is zero over a part of the boundary, and Ω is connected, we have $v = 0$ in Ω. Thus, $v_{\sigma(m)} \to 0$ in H^1, which contradicts $\|v_{\sigma(m)}\|_{H^1} = 1$.

3.4 Stampacchia Theorem and application

The Stampacchia theorem (Guido Stampacchia, Italian mathematician, Naples, March 26, 1922 – Paris, April 22, 1978) is a simple yet powerful extension of the Lax-Milgram result to closed convex subsets of Hilbert spaces.

Theorem 3.2. Let V be an Hilbert space, let a be a bilinear form on $V \times V$ and ℓ a linear form on V verifying the assumptions of Lax-Milgram (continuity, coerciveness). Let K be a

closed convex subset of V. Then there exists a unique $u \in K$ such that

$$\forall v \in K, \quad a(u, v-u) \geq \ell(v-u).$$

Moreover, if a est symmetric, then the above u is the unique minimizer on K of the quadratic form $\frac{1}{2}a(v,v) - \ell(v)$.

Proof. Arguing as in Lax-Milgram theorem we associate to the form a, thanks to Riesz theorem, an operator A. Likewise we associate to ℓ a vector b. We thus seek $u \in K$ such that $\langle b - Au, v-u\rangle \leq 0$ for all $v \in K$. This also writes for all $r > 0$, $\langle r(b-Au)+u-u, v-u\rangle \leq 0$. This could be think as a projection on K : u is such that for all r, $u = p_K(r(b-Au)+u)$, where p_K is the projection operator on K. We therefore seek a fixed point of $u \to p_K(r(b-Au)+u)$. As a projection operator is Lipschitzian with constant 1 (aka non-expansive), we just have to prove that $u \to r(b-Au)+u$ is contractant to finish the proof. This is exactly what we proved for the Lax-Milgram theorem. $\square$

We will see an application of Lax-Milgram theorem to non-homogeneous boundary problems in the next subsection. Another application is to obstacle problems. Let us consider the $1D$ problem of looking for the shape of the shortest inextensible rope linking to points (say $x = -1$ and $x = 1$) while remaining above some fixed obstacle. That is, find $u : (-1,1) \to \mathbb{R}$ such that $u(-1) = u(1) = 0$ and $u \geq g$ where $g \in C^2([-1,1])$ is such that $g(-1) = g(1) \leq 0$, and minimizing its length

$$\int_{-1}^{1} \sqrt{1+u'(x)^2}dx$$

Writing $\sqrt{1+u'(x)^2} \approx 1 + \frac{1}{2}u'(x)^2$, we replace this minimization by the simpler one on:

$$\int_{-1}^{1} u'(x)^2 dx$$

This problem could be cast under the Stampacchia form: We set $V = H_0^1(-1,1)$, $K = \{u \in V,\ u \geq g\}$ and $a(u,v) = \int_{-1}^{1} u'v'dx$. Then the former theorem states that there is a unique $u \in K$ such that

$$\int_{-1}^{1} u'(v'-u')dx \geq 0, \qquad \forall v \in K$$

Intuitively, for a concave g, on subsets of $(-1,1)$ where $u > g$, then we can take v below or above u, thus the variational inequality above leads to $u'' = 0$, that is, the graph of u is a straight line. While when $u = g$ we know already g. We can check that for $g(x) = 1 - 2x^2$ we obtain the left situation of figure 3.1. Indeed for this solution, is we denote by x_0 and x_1 the two points

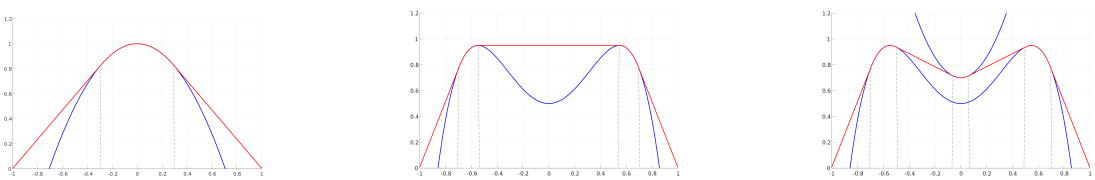


Figure 3.1: Simple concave obstacle (Left), non concave obstacle (Center), and double obstacle (Right)

of contact, then since u'' cancels outside (x_0, x_1), and coincide with g on that interval, we have by integration by parts,

$$\int_{-1}^{1} u'(v'-u')dx = \int_{x_0}^{x_1} u''(u-v)dx = \int_{x_0}^{x_1} g''(g-v)dx \geq 0$$

since g is concave, $g'' < 0$ and $g \in K$ which means $v \geq g$. By uniqueness we see that this is the unique solution. We can check likewise that the Stampacchia criterium is verified for the non concave case as well. For the last case, we add another obstacle h whose graph is convex and consider now the set $K = \{u \in V,\ h \geq u \geq g\}$, and with the same reasonning it is easy to show that the Stampacchia inequality is satisfied for the function correspondng to the right picture of figure 3.1.

This simple obstacle problem is used for pricing American call or put options in mathematical finance.

3.5 Maximum principles

We are going to prove that elliptic operators have properties which ensures that solution of Dirichlet or Neumann boundary problem satisfies to some maximum principle (or comparison principle). Let us first start with the (nonhomogeneous) Dirichlet problem on a bounded domain Ω:

$$(D) \qquad \begin{cases} -\Delta u + u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases}$$

where $f \in L^2(\Omega)$. We can cast the existence question for this problem in the Stampacchia framework: Let us consider $K = \{u \in H^1(\Omega),\ u = g \text{ on } \partial\Omega\}$ which is a closed convex (affine) subset of $H^1(\Omega)$. The closeness stems from the continuity of the trace map. Consider on V the bilinear/linear forms

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v + uvdx, \qquad \ell(v) = \int_\Omega fvdx,$$

which verify the Lax-Milgrams assumptions on V as already seen. Thus the Stampacchia theorem states that there exist a unique $u \in K$ such that

$$\forall v \in K, \quad a(u, v-u) \geq \ell(v-u).$$

In this particular case, as K is an affine subspace of V, while v spans K, $w = v - u$ spans $H_0^1(\Omega)$. From the (bi)linearity of a and ℓ, the inequality $a(u,w) \geq \ell(w)$ for all $w \in H_0^1(\Omega)$ turns to an equality. We therefore obtain the existence and uniqueness of a function $u \in K$ such that

$$\int_\Omega \nabla u \cdot \nabla v + uvdx = \int_\Omega fvdx, \qquad \forall v \in H_0^1(\Omega) \tag{3.1}$$

which leads to the first equation of (D) in the distribution and $L^2(\Omega)$ meaning.

Let us study some properties of this solution. The first result states that a continuous variational solution of (D) cannot take arbitrary values inside the domain:

Theorem 3.3. Let $u \in K \cap C^0(\overline{\Omega})$ verifying

$$\int_\Omega \nabla u \cdot \nabla v + uv dx = \int_\Omega f v dx, \qquad \forall v \in H_0^1(\Omega) \tag{3.2}$$

then for all $x \in \Omega$,

$$\min(\inf_\Omega f, \inf_{\partial\Omega} g) \leq u(x) \leq \max(\sup_\Omega f, \sup_{\partial\Omega} g)$$

Remark 3.4.

1. In the case of homogeneous Dirichlet problem, $\inf_{\partial\Omega} g = \sup_{\partial\Omega} g = 0$ thus we end up with the maximum principle

$$\min(\inf_\Omega f, 0) \leq u(x) \leq \max(\sup_\Omega f, 0)$$

2. If $f = 0$ then u is called harmonic, and this maximum principle reads

$$\min(0, \inf_{\partial\Omega} g) \leq u(x) \leq \max(0, \sup_{\partial\Omega} g)$$

which means that such a function cannot reach an extremum inside Ω. In the 1D case, this is trivial : an affine function takes its min and max on the boundary.

Proof. This proof is interesting since it can be applied to many situations and uses an important tool: the Stampacchia truncation method. Let consider an increasing function $G \in C^1(\mathbb{R})$ such that G' is bounded on $\mathbb{R}$ by some constant M, and $G = 0$ on $\mathbb{R}^-$. Then we set $K = \max(\sup_\Omega f, \sup_{\partial\Omega} g)$ and v=G(u-K). Then $v \in H^1(\Omega)$. Indeed since $G(r) \leq Mr$, $v \in L^2(\Omega)$, and since $G' \leq M$, $\nabla v = G'(u-K)\nabla u \in L^2(\Omega)$. Moreover, $v \in H_0^1(\Omega)$ since $u = g$ implies $v = G(g-K) = 0$, since $g - K \leq 0$. Then using v in the variational formulation leads to

$$\int_\Omega G'(u-K)|\nabla u|^2 + uG(u-K)dx = \int_\Omega fG(u-K)dx$$

Subtracting $KG(u-K)$ on both sides gives

$$\int_\Omega G'(u-K)|\nabla u|^2 + (u-K)G(u-K)dx = \int_\Omega (f-K)G(u-K)dx$$

From the assumptions, the first term is nonnegative, and the right hand side is less or equal to zero since $f \leq K$ and $G \geq 0$. Thus

$$\int_\Omega (u-K)G(u-K)dx \leq 0$$

Since $tG(t) \geq 0$, we get $(u-K)G(u-K) = 0$ on Ω. But still from the assumption on G, this can happen if and only if $u \leq K$. □

Corollary 3.1. Let $u \in H^1(\Omega) \cap C^0(\overline{\Omega})$ verifying (3.2) then

$$[u \geq 0 \text{ on } \partial\Omega \text{ and } f \geq 0 \text{ in } \Omega] \Longrightarrow [u \geq 0 \text{ in } \Omega],$$

$$\|u\|_{L^\infty(\Omega)} \leq \max(\|u\|_{L^\infty(\partial\Omega)}, \|f\|_{L^\infty(\Omega)}]$$

This corollary extend to general elliptic operators, and to Neumann boundary value problems (see [4] for an extensive study).

Chapter 4

Variational approximation of elliptic problems

We want to solve in an approximate way the formulation: find $u \in V$, $\forall v \in V$, $a(u,v) = \ell(v)$ with a and ℓ verifying the hypotheses of Lax-Milgram's theorem.

4.1 Inner approximation

Let $V_h \subset V$ be a finite-dimensional vectorial subspace, such as $V_h \to V$ when $h \to 0$, with a meaning that we will specify later. We consider the approximated problem: find $u_h \in V_h$, $\forall v_h \in V_h$, $a(u_h, v_h) = \ell(v_h)$. This way of approaching the variational problem is called inner, because we remain in a subspace of the continuous space V, unlike the finite difference method that leaves the domain of regularity of the solutions of the equations it approaches. We denote by (FV) the continuous variational formulation, and (FV_h) the approximate variational formulation.

Proposition 4.1. (FV_h) has a unique solution.

Proof. A quick proof would be to apply Lax-Milgram on V_h which is a Hilbert space since of finite dimension. More interestingly, we solve this problem by hand: let $(\phi_1, \dots, \phi_n)$ be a basis of V_h. Find u_h corresponds to finding its components on this basis, that is $u_h = \sum_{j=1}^n u_h^j \phi_j$, so that the problem is to find the real numbers (u_h^j), such as

$$\sum_{j=1}^n a(\phi_j, \phi_i) u_h^j = \ell(\phi_i)$$

for $i = 1, \dots, n$. This corresponds to a linear system of matrix $(a(\phi_j, \phi_i))$. This one is invertible because the solution $\xi \in \mathbb{R}^n$ of the homogeneous system verifies

$$\sum_{j=1}^n a(\phi_j, \phi_i) \xi^j = 0.$$

It gives

$$\sum_{i,j=1}^n a(\phi_j, \phi_i) \xi^j \xi^i = 0$$

so $\alpha\|\xi\|^2 = 0$ under the Lax-Milgram theorem assumptions. □

The success of variational methods (or inner approximation) is based in particular on the simplicity of the following result.

Theorem 4.1 (Céa). Let u solution of (FV) and u_h solution of (FV_h). Then there exists $C > 0$ independent of h such that $\|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V$.

The problem of convergence of the method for the elliptic equations is thus reduced to a problem of approximation of the functions of V by functions of V_h.

Proof. Let $v_h \in V_h$, and $w_h = v_h - u_h \in V_h \subset V$. Since w_h is in V and in V_h, it is a valid test function in (FV) and (FV_h). We thus have

$$a(u_h, w_h) = \ell(w_h) \quad \text{and} \quad a(u, w_h) = \ell(w_h),$$

so $a(u - u_h, w_h) = 0$. By coercivity and continuity of a, we have

$$\alpha\|u - u_h\|^2 \leq a(u - u_h, u - u_h) = a(u - u_h, u - v_h) + a(u - u_h, w_h) \leq M\|u - u_h\|\|u - v_h\|.$$

So $\|u - u_h\| \leq \frac{M}{\alpha}\|u - v_h\|$, and this for all v_h, hence the result. □

We've been saying for a while $V_h \to V$ when $h \to 0$. Here's what we need in order that the method converges.

Theorem 4.2. Let $\mathcal{V}$ be a dense subspace of V and $r_h : \mathcal{V} \to V_h$ such that $\forall v \in \mathcal{V}$, $\lim_{h\to 0} \|v - r_h v\|_V = 0$. Then the variational approximation method converges: the solution u_h of (FV_h) tends to the u solution of (FV) when $h \to 0$.

Proof. Indeed, let u be the solution of (FV). Let $\varepsilon > 0$, by density of $\mathcal{V}$ in V there exists $u_\varepsilon \in \mathcal{V}$ such that $\|u - u_\varepsilon\|_V \leq \frac{\varepsilon}{2}$. On the other hand, since $u_\varepsilon \in \mathcal{V}$ we have by hypothesis $\exists h_0 > 0$, such that for $h < h_0$ we have $\|u_\varepsilon - r_h u_\varepsilon\| \leq \frac{\varepsilon}{2}$. So

$$\inf_{v_h \in V_h} \|u - v_h\| \leq \inf_{v_h \in V_h} \|u_\varepsilon - v_h\| + \|u - u_\varepsilon\| \leq \|u_\varepsilon - r_h u_\varepsilon\| + \frac{\varepsilon}{2} \leq C\varepsilon.$$

According to the previous theorem, we have proved that for $\varepsilon > 0$ fixed,

$$\exists h_0 > 0,\ h < h_0, \implies \|u - u_h\| \leq C\varepsilon$$

which is the desired convergence result. □

Converging is one thing, but you still need to know how fast. This depends, according to the above theorem, on the quality of the approximation of V by V_h. This is how the different methods of finite elements are distinguished. Let's first look at the theoretical construction of a space V_h.

4.2 Ritz-Galerkin and Petrov-Galerkin methods

4.2.1 Hilbertian basis

An Hilbertian basis of an Hilbert space V is an orthonormal countable family which spans all V. If V is a separable Hilbert space, that is there exists a countable dense family in V, we can build infinitely many Hilbertian basis by the following process. Left to take a dense free subfamily, let's assume the dense family free. The vectors of this family are taken successively and orthonormalization is carried out by the Gram-Schmidt method. We then get an Hilbertian basis $(w_1, w_2, \ldots, w_m, \ldots)$.

4.2.2 Choice of approximation space

A Galerkin method consists in choosing one of these Hilbertian families, and for $h = \frac{1}{m}$, $V_h = \mathrm{vect}(w_1, w_2, \ldots, w_m)$. We therefore try to solve: find $u_h \in V_h$, $\forall v_h \in V_h$, $a(u_h, v_h) = \ell(v_h)$.

Theorem 4.3. For any integer $m \in \mathbb{N}$, there is a unique solution $u_m \in V_{\frac{1}{m}}$ and we have

$$\lim_{m\to\infty} \|u - u_m\| = 0$$

where u is the solution of the original (non approximated) problem.

Proof. We consider for r_h the projector on V_h, and for $\mathcal{V}$ of the former theorem, the V space itself. Then, according to the density of the Hilbertian family in V, we have $\forall v \in \mathcal{V}$, $\lim_{h\to 0} \|v - r_h v\|_V = 0$ which guarantees the convergence of the method. □

Example:

1. Consider first the following simple problem: $-u'' + u = 1$ on $]0, 1[$, $u(0) = 0$ and $u'(1) = -1$. We therefore consider the space $V = \{u \in H^1,\ u(0) = 0\}$ and we approximate it by $V_m = \mathrm{span}\left((x^j)_{1\le j\le m}\right)$ for a given $m \in \mathbb{N}^*$. The approximate variational formulation is: Find $u \in V_m$ such that for all $i = 1 \ldots m$,

$$\int_0^1 \left(u'(x) i x^{i-1} + u(x) x^i\right) dx - u'(1) 1^i + 0 = \int_0^1 x^i dx.$$

Given the boundary condition in $x = 1$ and using the decomposition $u = \sum_{j=1}^m u_j x^j$, we obtain

$$\sum_{j=1}^m u_j \int_0^1 \left(ij x^{i+j-2} + x^{i+j}\right) dx = -1 + \int_0^1 x^i dx.$$

Finally we get the linear system:

$$\sum_{j=1}^m \left(\frac{ij}{i+j-1} + \frac{1}{i+j+1}\right) u_j = -1 + \frac{1}{i+1}, \quad \text{for all } i = 1 \ldots m.$$

Concerning the exact solution, a small calculation gives $u_e(x) = 1 - \mathrm{ch}\, x + \frac{-1+\mathrm{sh}\,1}{\mathrm{ch}\,1} \mathrm{sh}\, x$. The curves of the analytical solution and the approximate solution for some values of m are represented 4.1. There is a convergence when m increases. However for $m = 7$, the condition number of the associated matrix is of the order of 10^8 on this very simple case, and beyond a loss of accuracy occurs!

2. We now consider a two-dimensional problem on the unit square $\Omega =]0, 1[\times]0, 1[$ with $u = 0$ as boundary conditions on the boundary parts $x = 0$ and $y = 0$, and respectively equal to x and y on the boundary parts $y = 1$ and $x = 1$. Then we consider the problem: $-\Delta u = 10$ in Ω. To get back to a problem with null data at the boundary, we set $v(x, y) = u(x, y) - xy$ which is a solution of the same PDE in Ω. We choose as the basis the family of function $\phi_i = \sin(i_1 \pi x) \sin(i_2 \pi y)$, for $i_1, i_2 = 1, \ldots, m$, which we order in the following way (example from A. Fortin and A. Garon): $\phi_1 = \sin \pi x \sin \pi y$, $\phi_2 = \sin \pi x \sin 2\pi y$,

Table 4.1: Convergence table for Ritz-Galerkin method in dimension 1

m	$\|u_e - u_m\|_\infty / \|u_e\|_\infty$
1	0.2909501
2	0.0077572
3	0.0013430
4	0.0000246
5	0.0000027
6	3.539D-08

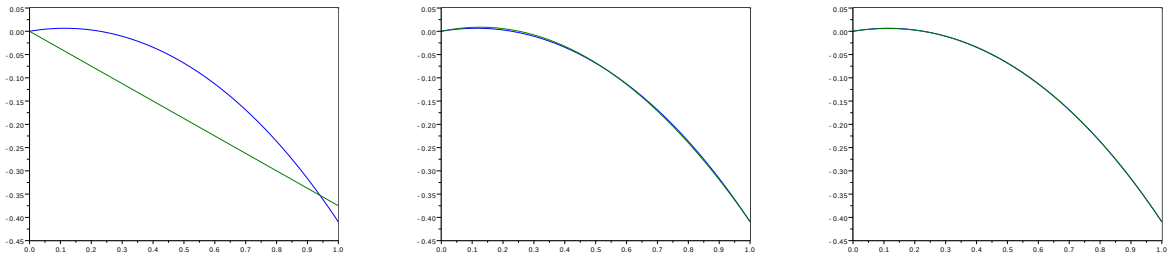


Figure 4.1: Convergence of Ritz-Galerkin method, for $m = 1, 2, 3$ toward the exact solution.

..., $\phi_m = \sin \pi x \sin m\pi y$, $\phi_{m+1} = \sin 2\pi x \sin \pi y$, ..., $\phi_{2m} = \sin 2\pi x \sin m\pi y$, ..., $\phi_{m^2} = \sin m\pi x \sin m\pi y$. The general formula gives the index of the function according to the modes of the sines is $i = (i_1 - 1)m + i_2$. These functions indeed cancel out on the boundary of Ω and on the other hand are orthogonal for the scalar product of H_0^1. We indeed have $a_{ij} = \frac{\pi^2}{4}(i_1^2 + i_2^2)\delta_{ij}$. So the system to solve in the variational formulation is diagonal and the second member is simply

$$10 \int_\Omega \phi_i(x, y) dxdy = \frac{10}{\pi^2} \left(\frac{(-1)^{i_1} - 1}{i_1} \right) \left(\frac{(-1)^{i_2} - 1}{i_2} \right).$$

So we have the Ritz-Galerkin solution in an explicit way by:

$$u_m(x, y) = xy + \frac{40}{\pi^4} \sum_{i_1, i_2 = 1}^{m} \frac{1}{i_1^2 + i_2^2} \left(\frac{(-1)^{i_1} - 1}{i_1} \right) \left(\frac{(-1)^{i_2} - 1}{i_2} \right) \sin(i_1 \pi x) \sin(i_2 \pi y).$$

Remarques

1. If we choose for u_h the same basis of V_h as for v_h, the method is called Ritz-Galerkin's method. Otherwise we talk about Petrov-Galerkin's method. In this case we introduce two bases such as $V_h = \text{vect}(w_1, w_2, \ldots, w_m) = \text{vect}(\tilde{w}_1, \tilde{w}_2, \ldots, \tilde{w}_m)$, and the system to solve is $\sum_{j=1}^m u_j a(w_j, \tilde{w}_i) = \ell(\tilde{w}_i)$ for $i = 1, \ldots, m$. The interest of choosing two different bases occurs during the resolution of some particular PDEs (convection-diffusion equations).

2. This method is mostly used theoretically in the proof of existence of PDE solutions. It allows to reduce a problem of infinite dimension to a linear system of finite size. In practice, however, the method is not used as it was just presented, because the conditioning of the matrix obtained is rather bad and it is often full. On the other hand, the determination of basis functions verifying the boundary conditions can be very complicated for non-academic cases.

Rather, we will consider an approximate space V_h based on a discretization of the space domain and described by a basis (ϕ_i) chosen such that the terms $a(\phi_j, \phi_i)$ are as often as possible null.

4.3 Finite elements: example in dimension 1

4.3.1 Toy problem

We consider the problem of Dirichlet on the domain $\Omega =]0,1[$: Find $u : \Omega \to \mathbb{R}$ solution of

$$\begin{cases} -u'' + u = f & \text{on }]0,1[\\ u(0) = u(1) = 0 \end{cases}$$

The Ω domain is discretized by subdividing $(x_j)_{0 \leq j \leq n+1}$ with $x_j = jh$, $h = \frac{1}{n+1}$.

4.3.2 $\mathbb{P}_1$ finite elements

We consider as a set V_h:

$$V_h = \left\{ u \in C^0([0,1]), \quad \forall i = 0, \dots n, \ u_{|[x_i, x_{i+1}]} \in \mathcal{P}_1 \right\}$$

where $\mathcal{P}_1$ is the set of polynomials of degree less or equal to 1 on $\mathbb{R}$. Likewise, we set

$$V_h^0 = \{ u \in V_h, \quad u(0) = u(1) = 0 \}$$

According to the proposition 2.2, these two spaces are subspaces of $H^1(0,1)$, therefore Hilbert spaces since they are of finite dimension. Indeed, by introducing the $\phi : \mathbb{R} \to \mathbb{R}$ function defined by $\phi(x) = 1 - |x|$ over $[-1,1]$ and anywhere else, a base of V_h is formed of the functions $\phi_j(x) = \phi(\frac{x - x_j}{h})$, for $j = 0, \dots, n+1$. They are called the basic functions for finite elements $\mathbb{P}_1$. Likewise V_h^0 is generated by these same functions for $j = 1, \dots, n$. The spaces V_h and V_h^0 are of dimension $n+2$ and n respectively. An element $u_h \in V_h$ is expressed in the form $u_h(x) = \sum_{j=0}^{n+1} u_h(x_j)\phi_j(x)$ while $u_h^0 \in V_h^0$ will have the expression $u_h^0(x) = \sum_{j=1}^{n} u_h(x_j)\phi_j(x)$. The approximate variational formulation is written for this Dirichlet problem on V_h^0 to transcribe the boundary conditions. The variational formulation being of the form $a(u,v) = \ell(v)$ with $a(u,v) = \int_0^1 u'v' + uv dx$ and $\ell(v) = \int_0^1 fv dx$, we get by replacing u_h by its expression in the basis ϕ_j the problem: find $(u_j)_{1 \leq j \leq n} \in \mathbb{R}^n$ such that

$$\sum_{j=1}^{n} a(\phi_j, \phi_i) u_j = \ell(\phi_i), \qquad i = 1 \dots, n$$

which constitutes a system of n equations with n unknowns, with matrix coefficients $a(\phi_i, \phi_j)$ (a is symmetric). Let's compute these coefficients for the basis functions of $\mathbb{P}_1$. Since the support of ϕ_i is $[x_{i-1}, x_{i+1}]$, the product $\phi_i\phi_j$ is identically zero when $|i-j| > 1$. So we only have to compute $a(i, i-1)$, $a(i, i)$ and $a(i, i+1)$ knowing that by symmetry we have $a(i, i-1) = a(i, i+1)$. Let's compute:

$$a(\phi_i, \phi_i) = \frac{1}{h}\int_{x_{i-1}}^{x_{i+1}} \phi(\frac{x-x_i}{h})^2 dx + 2\int_{x_i}^{x_{i+1}} \phi(\frac{x-x_i}{h})^2 dx$$
$$= \frac{2}{h}\int_{x_i}^{x_{i+1}} 1^2 dx + 2\int_{x_i}^{x_{i+1}} \left(1 - (\frac{x-x_i}{h})\right)^2 dx.$$

We find $a(\phi_i, \phi_i) = \frac{2}{h} + \frac{2h}{3}$. An analogous calculation gives $a(\phi_i, \phi_{i+1}) = -\frac{1}{h} + \frac{h}{6}$. The system to be solved is therefore tridiagonal. To go further in the study of this method, we need an approximation result of functions of $H^1(0,1)$ by elements of V_h:

Theorem 4.4. Let r_h be the interpolation operator from $H^1(0,1)$ to V_h. The for all $u \in H^1(0,1)$, $\lim_{h\to 0} \|r_h u - u\|_{H^1(0,1)} = 0$ and there exists $C > 0$ such that for all $u \in H^2(0,1)$, $\|r_h u - u\|_{H^1(0,1)} \leq Ch\|u''\|_{L^2(0,1)}$.

Corollary 4.1. Let $u \in H_0^1(\Omega)$ be solution of (FV) and $u_h \in V_h^0$ solution of (FV_h). Then the $\mathbb{P}_1$ finite element method converges: $\lim_{h\to 0} \|u_h - u\|_{H^1(0,1)} = 0$ and there exists $C > 0$ such that if $u \in H^2(0,1)$ we have $\|u - u_h\| \leq Ch\|u''\|_{L^2(0,1)}$.

Proof. On $[x_i, x_{i+1}]$ we have for $v \in \mathcal{C}^\infty([0,1])$, $r_h v(x) = v(x_i) + \frac{x-x_i}{x_{i+1}-x_i}(v(x_{i+1}) - v(x_i)) \in \mathcal{P}_1$. Let us write a first estimate. We have

$$\begin{aligned} r_h v(x) - v(x) &= v(x_i) - v(x) + \frac{x-x_i}{x_{i+1}-x_i}(v(x_{i+1}) - v(x_i)) \\ &= \int_x^{x_i} v'(t)dt + \frac{x-x_i}{x_{i+1}-x_i}\int_{x_i}^{x_{i+1}} v'(t)dt \\ &= (x_i - x)v'(\xi_1) + (x - x_i)v'(\xi_2) = (x - x_i)(v'(\xi_2) - v'(\xi_1) \\ &= (x - x_i)\int_{\xi_1}^{\xi_2} v''(t)dt \end{aligned}$$

Thus

$$\begin{aligned} (r_h v(x) - v(x))^2 &\leq (x - x_i)^2 \left(\int_{x_i}^{x_{i+1}} |v''(t)|dt\right)^2 \\ &\leq h^2 \left(\int_{x_i}^{x_{i+1}} 1^2 dt\right) \|v''\|^2_{L^2(x_i,x_{i+1})} \leq h^3\|v''\|^2_{L^2(x_i,x_{i+1})}. \end{aligned}$$

Integrating on $[x_i, x_{i+1}]$ we get $\int_{x_i}^{x_{i+1}} (r_h v(x) - v(x))^2 dx \leq h^4\|v''\|^2_{L^2(x_i,x_{i+1})}$ and finally

$$\|r_h v - v\|_{L^2(0,1)} \leq h^2\|v''\|_{L^2(0,1)}.$$

This inequality is valid in $H^2(0,1)$ by a density argument. Let us now turn to the derivatives estimates:

$$(r_h v)' - v'(x) = \frac{v(x_{i+1}) - v(x_i)}{x_{i+1} - x_i} - v'(x) = \frac{1}{h}\int_{x_i}^{x_{i+1}} v'(t) - v'(x)dt = \frac{1}{h}\int_{x_i}^{x_{i+1}}\int_x^t v''(y)dydt$$

thus

$$\begin{aligned}|(r_h v)' - v'(x)|^2 &\leq \frac{1}{h^2}\left(\int_{x_i}^{x_{i+1}}\int_x^t v''(y)dydt\right)^2 \\ &\leq \frac{1}{h^2}\left(\int_{x_i}^{x_{i+1}} 1^2 dt\right)\left(\int_{x_i}^{x_{i+1}}\left(\int_x^t v''(y)dy\right)^2 dt\right) \\ &\leq \frac{1}{h}\int_{x_i}^{x_{i+1}}\int_x^t 1^2 dy \int_x^t v''^2(y)dydt \leq h\|v''\|^2_{L^2(x_i,x_{i+1})}\end{aligned}$$

Integrating on $[x_i, x_{i+1}]$ and summing on i we finally obtain

$$\|(r_h v)' - v'\|_{L^2(0,1)} \leq h\|v''\|_{L^2(0,1)}.$$

and this inequality is valid on $H^2(0,1)$ by density. By gathering these two inequalities we get:

$$\|r_h v - v\|_{H^1(0,1)} \leq Ch\|v''\|_{L^2(0,1)}.$$

We now prove the convergence result. Starting over on the estimate of the first order derivative,

$$r_h v - v(x) = \int_x^{x_i} v'(t)dt + \frac{x - x_i}{x_{i+1} - x_i}\int_{x_i}^{x_{i+1}} v'(t)dt$$

which gives

$$|r_h v - v(x)| \leq 2\int_{x_i}^{x_{i+1}} |v'(t)|dt$$

Thus thanks to Cauchy-Schwarz inequality,

$$|r_h v - v(x)|^2 \leq 4h\|v'\|^2_{L^2(x_i,x_{i+1})}$$

which upon integrating on $[x_i, x_{i+1}]$ gives

$$\|r_h v - v\|^2_{L^2(x_i,x_{i+1})} \leq 4h^2\|v'\|^2_{L^2(x_i,x_{i+1})}.$$

Finally, summing on i, we get

$$\|r_h v - v\|_{L^2(0,1)} \leq 2h\|v'\|_{L^2(0,1)}$$

To end up the proof, we have to prove that $\|(r_h v)' - v'\|_{L^2(0,1)} \to 0$. To that aim, we pick an $\varepsilon > 0$. As $\mathcal{C}^\infty([0,1])$ is densely imbedded in $H^1(0,1)$ there exists $\phi \in \mathcal{C}^\infty([0,1])$ such that $\|\phi - v\|_{H^1(0,1)} < \varepsilon$, and *a fortiori*, $\|\phi' - v'\|_{L^2(0,1)} < \varepsilon$. As $\phi \in \mathcal{C}^\infty([0,1]) \subset H^2(0,1)$, we can use the formerly proved inequality, $\|(r_h\phi)' - \phi'\|_{L^2(0,1)} \leq h\|\phi''\|_{L^2(0,1)}$. The we can get an estimation on $\|(r_h v)' - (r_h\phi)'\|_{L^2(0,1)}$ by writing in $[x_i, x_{i+1}]$

$$(r_h v)'(x) = \frac{v(x_{i+1}) - v(x_i)}{h} = \frac{1}{h}\int_{x_i}^{x_{i+1}} v'(t)dt$$

which gives $(r_h v)'(x)^2 \leq \frac{1}{h}\|v'\|_{L^2(x_i,x_{i+1})}$, and then by integrating on $[x_i, x_{i+1}]$,

$$\|(r_h v)'\|^2_{L^2(x_i,x_{i+1})} \leq \|v'\|^2_{L^2(x_i,x_{i+1})}$$

and at last by summing up on i,

$$\|(r_h v)'\|_{L^2(0,1)} \leq \|v'\|_{L^2(0,1)}$$

As a special choice of v we have thus

$$\|(r_h v)' - (r_h \phi)'\|_{L^2(0,1)} \leq \|v' - \phi'\|_{L^2(0,1)} < \varepsilon.$$

This inequality ends up the proof since

$$\|(r_h v)' - v'\|_{L^2(0,1)} \leq \|(r_h v)' - (r_h \phi)'\|_{L^2(0,1)} + \|(r_h \phi)' - \phi'\|_{L^2(0,1)} + \|\phi' - v'\|_{L^2(0,1)} \leq 2\varepsilon + h\|\phi''\|_{L^2(0,1)}$$

Let us pick $h_0 \leq \frac{\varepsilon}{\|\phi''\|_{L^2(0,1)}}$, then $h < h_0$ implies $\|(r_h v)' - v'\|_{L^2(0,1)} \leq 3\varepsilon$. Which is the desired convergence. □

4.3.3 $\mathbb{P}_2$ finite element

We now consider as set V_h:

$$V_h = \left\{u \in C^0([0,1]), \quad \forall i = 0, \dots n, \ u_{|[x_i, x_{i+1}]} \in \mathcal{P}_2\right\}$$

where $\mathcal{P}_2$ is the set of polynomial functions on $\mathbb{R}$,of degree less or equal to 2. As well we set

$$V_h^0 = \{u \in V_h, \quad u(0) = u(1) = 0\}$$

To build up a basis for V_h, we consider as in $\mathbb{P}_1$ the Lagrange polynomials on $[x_i, x_{i+1}]$. These are

$$\psi_i(x) = \frac{2}{h^2}(x - x_{i+\frac{1}{2}})(x - x_{i+1}), \quad \psi_{i+1}(x) = \frac{2}{h^2}(x - x_{i+\frac{1}{2}})(x - x_i), \quad \psi_{i+\frac{1}{2}}(x) = -\frac{4}{h^2}(x - x_i)(x - x_{i+1})$$

The support set of ψ_i is $[x_{i-1}, x_{i+1}]$ and those of $\psi_{i+\frac{1}{2}}$ the interval $[x_i, x_{i+1}]$. One element of V_h writes

$$v(x) = \sum_{i=0}^{n+1} v(x_i)\psi_i(x) + \sum_{i=0}^{n} v(x_{i+\frac{1}{2}})\psi_{i+\frac{1}{2}}(x)$$

which means that this space is of dimension $2n + 3$. Likewise an element of V_h^0 writes

$$v(x) = \sum_{i=1}^{n} v(x_i)\psi_i(x) + \sum_{i=0}^{n} v(x_{i+\frac{1}{2}})\psi_{i+\frac{1}{2}}(x)$$

which means that this space is of dimension $2n + 1$. For the Dirichlet problem associated to the differential equation $-u'' = f$ sur $[0, 1]$, we therefore look for a solution

$$u_h = \sum_{k=1}^{2n+1} u_{k/2}\psi_{k/2}(x)$$

and the problem boils down to solve the linear system $K_h U_h = F_h$ where $K_h^{k\ell} = \int_0^1 \psi'_{k/2}(x)\psi'_{\ell/2}$. The linear system matrix is given by:

$$K_h = \frac{1}{h}\begin{pmatrix} \frac{16}{3} & -\frac{8}{3} & 0 & 0 & 0 & \cdots \\ -\frac{8}{3} & \frac{14}{3} & -\frac{8}{3} & \frac{1}{3} & 0 & \cdots \\ 0 & -\frac{8}{3} & \frac{16}{3} & -\frac{8}{3} & 0 & \cdots \\ 0 & \frac{1}{3} & -\frac{8}{3} & \frac{16}{3} & -\frac{8}{3} & \frac{1}{3} \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots \end{pmatrix}$$

It is therefore a little less sparse than in the case of $\mathbb{P}_1$ approximation, and twice as large. What is the interest of this approximation? It holds in the following convergence result.

Theorem 4.5. Let $u \in H_0^1(0,1)$ and $u_h \in V_h^0$ solution of the exact and approximate variational formulations associated with the problem above. Then the $\mathbb{P}_2$ method converges and $\lim_{h\to 0} \|u - u_h\|_{H^1(0,1)} = 0$. Moreover, if $u \in H^3(0,1)$, then there exists $C > 0$ independent of h such that

$$\|u - u_h\|_{H^1(0,1)} \leq Ch^2 \|u''\|_{L^2(0,1)}.$$

So the $\mathbb{P}_2$ method has quadratic convergence in h for regular solutions. It's profitable for n big enough. Indeed if we consider that the cost of resolution of a linear system of size n and bandwidth b is of the order of $b \times n^2$, it will be of the order of $3n^2$ for the $\mathbb{P}_1$ and $5n^2$ method for $\mathbb{P}_2$. But to increase the precision by a factor of 100 for the first method, it will cost 100 times more, for only 10 times more for the second.

4.3.4 An original finite element

One of the pros of the finite element method is that it does not limit itself to defining degrees of freedom as values of some function or derivative at nodes as in the finite difference method. For instance, consider the variational problem on $V = H^1(\Omega)$ defined by the bilinear form $a(u,v) = \int_0^1 u'v'dx + \left(\int_0^1 udx\right)\left(\int_0^1 vdx\right)$ and the linear form $\ell(v) = \int_0^1 fv$ dx. The problem of finding $u \in V$ such that for every $v \in V$, $a(u,v) = \ell(v)$ admits a unique solution because it enters the hypotheses of Lax-Milgram's theorem. The continuity of a and ℓ is not a problem. For coerciveness, however, we have to show that there exists a constant $C > 0$ such that for all $u \in V$, $a(u,u) \geq C\|u\|_V^2$. To do this, proceed by the absurd assuming that, for any constant $C = \frac{1}{n}$, there is a function u_n such that $a(u_n,u_n) \leq \frac{1}{n}\|u_n\|_V^2$. Let $w_n = \frac{u_n}{\|u_n\|_V}$, which is such that $\|w_n\|_V = 1$. According to the bilinearity of a, we have $a(w_n,w_n) \leq \frac{1}{n}$ which results in $w_n' \to 0$ in L^2 and $\int_0^1 w_n dx \to \int_0^1 wdx$. According to theorem 2.7 (Rellich theorem), there is a subsequence $w_{\sigma(n)}$ strongly convergent in L^2 to some function w. By continuity of the derivation in the sense of distributions, we have $w' = 0$, and likewise for the integral, $\int_0^1 wdx = 0$, thus $w = 0$. Moreover $\|w_{\sigma(n)}\|_V = 1$ and $w_{\sigma(n)}$ converges strongly to w in H^1. So we should have in particular $\|w\|_V = 1$, which is in contradiction with $w = 0$. Let us inspect to which boundary value problem this is associated. Let us point out that choosing $v = 1$ in the variational formulation gives $\int_0^1 udx = \int_0^1 fdx$. Moreover, the variational formulation can also be written as

$$\int_0^1 u'v'dx + \int_0^1 \left(\int_0^1 udx\right) vdx = \int_0^1 fvdx$$

which leads to the equation:

$$-u'' + \int_0^1 u dx = f \qquad u'(0) = u'(1) = 0 \qquad \int_0^1 u dx = \int_0^1 f dx.$$

We could use $\mathbb{P}_1$ finite elements to get an approximation of this problem. Let (ϕ_i) the associated finite element basis, the corresponding linear system to be solved would have element i, j equal to

$$\sum_j u_j \left(\int_0^1 \phi_i' \phi_j' + \left(\int_0^1 \phi_i dx \right) \left(\int_0^1 \phi_j dx \right) \right)$$

which is not sparse at all. Let us rather introduce an *ad hoc* finite element such that on the reference interval, the degrees of freedom are the values at both ends, plus the integral over the interval. To that aim we consider three polynomials on $[0, 1]$, ψ_0, ψ_1, ψ_2 such that $\psi_0(0) = 1$, $\psi_0(1) = 0$, $\int_0^1 \psi_0 dx = 0$; $\psi_1(0) = 0$, $\psi_1(1) = 0$, $\int_0^1 \psi_1 dx = 1$; $\psi_2(0) = 0$, $\psi_2(1) = 1$, $\int_0^1 \psi_2 dx = 0$, whose analytic expressions are:

$$\psi_0(x) = 3x^2 - 4x + 1 \qquad \psi_1(x) = 6x(1 - x) \qquad \psi_2(x) = 3x^2 - 2x.$$

We then define from ψ_1 the basis function $\phi_{i+\frac{1}{2}}$ with support in $[x_i, x_{i+1}]$ and from ψ_0 basis functions with support in $[x_{i-1}, x_{i+1}]$. As in the $\mathbb{P}_2$ approximation there is therefore three non-zero basis function on $[x_i, x_{i+1}]$: $\phi_i, \phi_{i+1/2}$, and ϕ_{i+1}. We then look for an approximation written as:

$$u = \sum_{i=1}^{2N+1} u_{i/2} \phi_{i/2}.$$

This leads to the following linear system to solve:

$$\sum_{j=1}^{2N+1} u_{j/2} \left(\int_0^1 \phi_i' \phi_j' + \left(\int_0^1 \phi_i dx \right) \left(\int_0^1 \phi_j dx \right) \right) = \int_0^1 f \phi_{i/2} dx$$

which is more sparse than the previous one, because for even i, the second term is completely null and we find a 3 bandwidth on these lines. Other examples of finite elements involve derivatives values of the unknown at some points. These are the Hermite type finite element. They are used in particular when we want the approaching functions to be more than piecewise continuous in pieces, for example piecewise differentiable. This is the case, for example, in the study of order 4 PDEs for which the space H^2 is involved.

4.4 Definition of a finite element

A finite element is a triple $(K, \mathcal{P}, \Sigma)$, where

(i) K in a compact subset $\mathbb{R}^N$ of non-void interior.

(ii) $\mathcal{P}$ finite dimensional vector space of functions from K to $\mathbb{R}$.

(iii) $\Sigma = \{\ell_1, \ell_2, \ldots, \ell_m\}$ is a set of m linear forms on $\mathcal{P}$.

Examples : Lagrange $1D$ finite elements

$\mathbb{P}_1$: $K = [0,1]$, $\mathcal{P} = \mathbb{R}_1[X]$, $\Sigma = \{p \to p(0), p \to p(1)\}$

$\mathbb{P}_2$: $K = [0,1]$, $\mathcal{P} = \mathbb{R}_2[X]$, $\Sigma = \{p \to p(0), p \to p(1/2), p \to p(1)\}$

The $1D$ finite element with an integral dof corresponds to

$\overline{\mathbb{P}_1}$: $K = [0,1]$, $\mathcal{P} = \mathbb{R}_2[X]$, $\Sigma = \left\{p \to p(0), p \to \int_0^1 p(x)dx, p \to p(1)\right\}$

Another $1D$ finite element may be defined by

$\widetilde{\mathbb{P}_2}$: $K = [0,1]$, $\mathcal{P} = \mathbb{R}_2[X]$, $\Sigma = \left\{p \to p(0), p \to p'(\frac{1}{2}), p \to p(1)\right\}$

The $2D$ Lagrange finite elements (see below) correspond to

$\mathbb{P}_1$: $K = \{(x,y), x \geq 0, y \geq 0, x+y \leq 1\}$, $\mathcal{P} = \langle 1, x, y\rangle$, $\Sigma = \{p \to p(0,0), p \to p(0,1), p \to p(1,0)\}$

$\mathbb{Q}_1$: $K = [0,1]^2$, $\mathcal{P} = \langle 1, x, y, xy\rangle$, $\Sigma = \{p \to p(0,0), p \to p(0,1), p \to p(1,0), p \to p(1,0)\}$

Definition 4.1 (Unisolvance). The finite element $(K, \mathcal{P}, \Sigma)$ is unisolvant provided that

$$\forall(\alpha_1, \ldots, \alpha_m) \in \mathbb{R}^m,\ \exists! p \in \mathcal{P},\ \forall j = 1 \ldots m,\ \ell_j(p) = \alpha_j$$

or equivalently the map $(\ell_1, \ldots, \ell_m)$ from $\mathcal{P}$ to $\mathbb{R}^m$ is one to one. In this event, we call **basis functions** of the finite element the m functions $\psi_i \in \mathcal{P}$, $i = 1, \ldots, m$ such that $\ell_j(\psi_i) = \delta_{ij}$, where δ_{ij} is the Kronecker symbol.

Remark 4.1. For a finite element to be unisolvant this is necessary that $\operatorname{card}\Sigma = \dim\mathcal{P}$. In that case proving the unisolvance boils down to find the basis functions or to prove that the only element of $\mathcal{P}$ which cancels every element of Σ is the identically zero.

Examples : All the above finite element are unisolvant (exercise), except $\widetilde{\mathbb{P}_2}$. Indeed for the latter $x(1-x)$ belongs to $\mathcal{P}$ and cancels ℓ_1, ℓ_2 and ℓ_3. Thus the kernel of the map (ℓ_1, ℓ_2, ℓ_3) from $\mathbb{R}_2[X]$ to $\mathbb{R}^3$ contains that polynomial. On the other hand it easy to see that if Σ is changes to $\left\{p \to p(0), p \to p'(\frac{1}{3}), p \to p(1)\right\}$ (for instance), then the finite element becomes unisolvant.

Obviously in dimension higher than one, in applications, we will have to deal with compact K which are transformed version of the elementary above situations. We therefore need the following simple proposition which could be proved for training:

Proposition 4.2. Let $\hat{K}$ a compact set of $\mathbb{R}^N$ and $(\hat{K}, \hat{\mathcal{P}}, \hat{\Sigma})$ an unisolvant finite element. Let $F : \mathbb{R}^N \to \mathbb{R}^N$ a one-to-one function. Define $(K, \mathcal{P}, \Sigma)$ as the finite element such that

$$K = F(\hat{K}), \quad \mathcal{P} = \left\{p : K \to \mathbb{R},\ p \circ F \in \hat{P}\right\} \quad \Sigma = \left\{\ell : \mathcal{P} \to \mathbb{R},\ \ell(p) = \hat{\ell}(p \circ F),\ \hat{\ell} \in \hat{\Sigma}\right\}$$

Then this finite element is unisolvant, and the basis functions of $(K, \mathcal{P}, \Sigma)$ are obtained by setting $\psi_i = \hat{\psi}_i \circ F^{-1}$ where $\hat{\psi}_i$ are the basis functions of $(\hat{K}, \hat{\mathcal{P}}, \hat{\Sigma})$.

4.5 Finite element of dimension 2

4.5.1 Triangulation of a polygonal domain

Definition 4.2. Let Ω be an open polygonal domain of $\mathbb{R}^2$. A triangulation of Ω is a collection $\mathcal{T}$ of triangles $(K_i)_{i=1\dots n}$ such that $K_i \subset \overline{\Omega}$ and $\Omega = \bigcup_{i=1}^n K_i$, and the intersection of two triangles of $\mathcal{T}$ is either empty, a vertice, of a full triangle edge. We denote for a triangle K, its diameter

$$\operatorname{diam}(K) = \max_{(x,y)\in K^2} \|x-y\|$$

and its roundness

$$\rho(K) = \max_{B\subset K,\ B \text{ ball of radius r}} (2r).$$

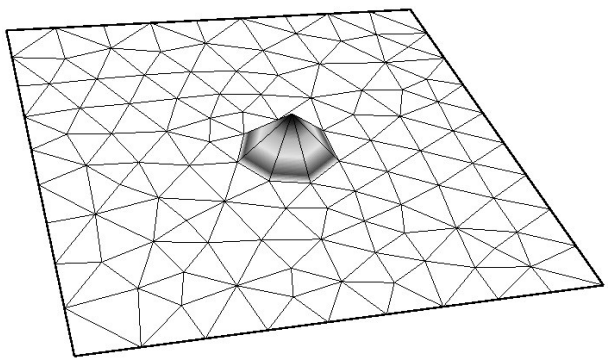
The ratio $\frac{\operatorname{diam}(K)}{\rho(K)}$ which is always ≥ 1 measures the flatness of K. A good mesh is such that the min of this ratio over the triangulation is not too small. This ratio is also involved in error estimate between a function and its interpolant on a given mesh.

4.5.2 $\mathbb{P}_1$ finite elements

Let the finite element space based on the triangulation described above,

$$V_h = \left\{v : \Omega \to \mathbb{R},\ v \in \mathcal{C}^0(\overline{\Omega}),\ v_{|K_i} \text{ affine}\right\}$$

A basis of this space, analogously with $\mathbb{P}_1$ element in dimension 1, is formed of the "hat" functions which achieve the value 1 on a given node of the triangulation, 0 on every other node and piecewise affine.



Barycentric coordinates: Let $K = (A_1, A_2, A_3)$ be a non flat triangle. We call barycentric coordinates of a given point M with respect to the triangle K the triplet $(\lambda_1, \lambda_2, \lambda_3)$ such that:

$$\begin{cases} \vec{OM} = \lambda_1 \vec{OA_1} + \lambda_2 \vec{OA_2} + \lambda_3 \vec{OA_3} \\ \lambda_1 + \lambda_2 + \lambda_3 = 1 \end{cases}$$

Those three numbers exists since the first equation could also be written as

$$\lambda_1 \vec{MA_1} + \lambda_2 \vec{MA_2} + \lambda_3 \vec{MA_3} = 0$$

and the vectors $\vec{MA_i}$ are linked in a two dimensional space.

For instance the triangle center has coordinates $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. We can verify that a point M belongs to the triangle if and only if its barycentric coordinates are all non-negative (and thus ≤ 1). A coordinate λ_i of a given point M is zero if an only if this point belongs to (A_jA_k) with $j, k \neq i$. More specifically, the middle of $[A_jA_k]$ as coordinate $\lambda_j = \lambda_k = \frac{1}{2}$ and thus $\lambda_i = 0$. At last, note that A_1 as coordinates $(1, 0, 0)$, A_2 $(0, 1, 0)$ and A_3 $(0, 0, 1)$.

The above definition leads to solve a linear system to find those barycentric coordinates. If M has cartesian coordinates (x, y) we solve:

$$\begin{pmatrix} x_{A_1} & x_{A_2} & x_{A_3} \\ y_{A_1} & y_{A_2} & y_{A_3} \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

As we see by solving this system, the λ_i functions are affine functions in (x, y). From the properties stated above, we see that they are just the Lagrange interpolating polynomials. We thus have:

$$\forall p \in \mathbb{P}_1, \quad p(x, y) = \sum_{i=1}^{3} p(A_i)\lambda_i(x, y).$$

We can also check that $\mathbb{P}_2$ polynomials on a triangle can be expressed in terms of those functions:

$$\forall p \in \mathbb{P}_2, \quad p(x, y) = \sum_{i=1}^{3} p(A_i)\lambda_i(x, y)(2\lambda_i(x, y) - 1) + 4 \sum_{1 \leq i < j \leq 3} p(A_{ij})\lambda_i(x, y)\lambda_j(x, y),$$

where A_{ij} is the middle of side $[A_iA_j]$.

4.5.3 Practical implementation

4.5.3.1 Toy example

Let $\Gamma_1 \cup \Gamma_2$ a partition of $\partial\Omega$. Let us consider the non-homogeneous Dirichlet problem

$$\begin{cases} -\Delta u + u = f & \text{on } \Omega \\ u = g & \text{on } \Gamma_1 \\ -\partial_n u + \alpha u = \phi & \text{on } \Gamma_2 \end{cases}$$

We define the spaces $V_0 = \left\{u \in H^1(\Omega),\ v_{|\Gamma_1} = 0\right\}$ and $V_g = \left\{u \in H^1(\Omega),\ v_{|\Gamma_1} = g\right\}$, the latter being an affine space directed by the former. Let $G_1 \in H^1(\Omega)$ such that $G_{1|\Gamma_1} = g$ (implying implicitely some regularity on g, since the range of the trace operator is not the whole L^2 space). We make the unknown change $U = u - G_1$, U which verifies

$$-\Delta U + U = -\Delta u + u + \Delta G_1 - G_1 = f + \Delta G_1 - G_1$$

in Ω and concerning the boundary

$$U_{|\Gamma_1} = 0, \qquad -\partial_n U + \alpha U_{|\Gamma_2} = -\partial_n u + \alpha u + \partial_n G_1 - \alpha G_1 = \phi + \partial_n G_1 - \alpha G_1.$$

We multiply by $v \in V_0$ and we integrate on Ω, to get

$$\int_\Omega \nabla U \nabla v + Uvdx - \int_{\Gamma_2} \partial_n U v d\sigma = \int_\Omega fv - \nabla G_1 \nabla v - G_1 v dx + \int_{\Gamma_2} \partial_n G_1 v d\sigma$$

since v cancels on Γ_1. Using the above boundary condition of U on Γ_2,

$$\int_\Omega \nabla U \nabla v + Uvdx - \int_{\Gamma_2} (\alpha U - \phi - \partial_n G_1 + \alpha G_1) v d\sigma = \int_\Omega fv - \nabla G_1 \nabla v - G_1 v dx + \int_{\Gamma_2} \partial_n G_1 v d\sigma$$

The normal derivatives terms of G_1 cancel and we recover a Lax-Milgram form $a(U, v) = \ell(v)$ by writing:

$$\int_\Omega \nabla U \nabla v + Uvdx - \alpha \int_{\Gamma_2} Uvd\sigma = \int_\Omega fv - \nabla G_1 \nabla v - G_1 v dx + \int_{\Gamma_2} (\phi - \alpha G_1) v d\sigma$$

We therefore get existence and uniqueness for the above variational formulation, and thus of the toy problem before transformation, since $u = U + G_1$.

Note however that $u \in V_g$ is solution to a variational problem set on V_0:

$$\int_\Omega \nabla u \nabla v + uvdx - \alpha \int_{\Gamma_2} uvd\sigma = \int_\Omega fvdx + \int_{\Gamma_2} \phi v d\sigma, \quad \forall v \in V_0$$

on which the Lax-Milgram theorem does not apply if $g \neq 0$. But one can apply directly the Stampacchia theorem on that situation.

4.5.3.2 Nodes numbering

We consider a $\mathbb{P}_1$ finite element approximation on a triangulation of Ω, which is assumed polygonal for that example. Let N_0 be the number of inner nodes of a triangulation of Ω, and I_0 the set of indices i of those points n_i. Likewise, let N_1 (resp. N_2) the number of nodes located on Γ_1 (resp. Γ_2), and I_1 (resp. I_2) the correponding sets of indices. From the boundary conditions of ou toy problem, an element $u_h \in V_h$ verifies $u_h(n_i) = 0$ for $i \in I_1$. The discrete weak formulation reads:

$$a(u_h, \phi_i) = \ell(\phi_i), \qquad \forall i \in I_0 \cup I_2$$

with for the bilinear form a, $G_1 = \sum_{i \in I_1} g(n_i)\phi_i(x)$. We thus end up with $N_0 + N_2$ linear equations on $N_0 + N_2$ unknowns. Note that G_1 only depends on the values of g on Γ_1.

4.5.3.3 Assembling

We must therefore compute, in the coefficients of the system, integrals of the form $\int_\Omega \phi_i \phi_j dx$ ou $\int_\Omega \nabla \phi_i \nabla \phi_j dx$. Since the basic functions are compactly supported, the most efficient way to do this is to go through all the elements (triangles) of the triangulation and compute which contribution this element will bring to the computation of the integral. Indeed, for instance, we can write

$$\int_\Omega \nabla \phi_i \nabla \phi_j dx = \sum_{K \in \mathcal{T}} \int_K \nabla \phi_i \nabla \phi_j dx$$

To compute the integral above on a triangle, we come back to a reference element by changing variables. Let's consider the element $\hat{K} = \{(\hat{x}, \hat{y}), \quad \hat{x} \geq 0, \hat{y} \geq 0, \hat{x} + \hat{y} \leq 1\}$ which is the elementary triangle of side 1 and vertices $A_1 = (0,0)$, $A_2 = (1,0)$, and $A_3 = (0,1)$. The $\mathbb{P}_1$ basis functions on that element are given by

$$\hat{\phi}_1(\hat{x}, \hat{y}) = 1 - \hat{x}\hat{y}, \quad \hat{\phi}_2(\hat{x}, \hat{y}) = \hat{x}, \quad \hat{\phi}_3(\hat{x}, \hat{y}) = \hat{y}.$$

The integral we want to compute is associated with a triangle K of vertices $A_1 = (x_1, y_1)$, $A_2 = (x_2, y_2)$, $A_3 = (x_3, y_3)$. We define an application $\mathcal{A}$ from $\hat{K}$ to K by mapping $\hat{M}(\hat{x}, \hat{y}) \in \hat{K}$ to $M(x, y) \in K$ in such a way that

$$\vec{A_1 M} = \hat{x}\vec{A_1 A_2} + \hat{y}\vec{A_1 A_3}.$$

We thus have

$$\begin{cases} x = \hat{x}(x_2 - x_1) + \hat{y}(x_3 - x_1) + x_1 \\ y = \hat{x}(y_2 - y_1) + \hat{y}(y_3 - y_1) + y_1 \end{cases}$$

The basic functions associated with K are obtained from those of $\hat{K}$ thanks to the application $\mathcal{A}$

$$\phi_i(x, y) = \hat{\phi}_i(\mathcal{A}^{-1}(x, y)).$$

Note that $\mathcal{A}$ is invertible, because it is composed of a translation and a linear application of determinant the area of the triangle K. To compute an integral on the triangle K we reduce to $\hat{K}$ by the change of variables $\mathcal{A}$. For a function of two variables F we have

$$\int_K F(x, y) dx dy = \int_{\hat{K}} F(\mathcal{A}(\hat{x}, \hat{y}))|J| d\hat{x} d\hat{y}$$

Chapter 5

Stokes equations

In this chapter, we consider the Stokes problem, a simplified model of fluid mechanics. Because of the incompressibility condition, the mathematical framework introduced in the previous chapters cannot be used directly. We will introduce a mixed variational formulation and then give details on the associated mixed finite element method.

5.1 Introduction

An incompressible viscous fluid flows at the speed $u(x, y)$ assumed to be constant in time. On a surface of normal n, the effort exerted is Σn where Σ is the stress tensor. For a Newtonian fluid it is written $\Sigma = -p\mathbb{I} + 2\mu D(u)$ with $D(u)_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j)$, where $U = (U_1, U_2)$. This matrix measures how the velocity varies between two neighboring particles, and the shape of the stress tensor reflects the fact that the fluid is incompressible and therefore exerts a normal force related to the pressure, and the fact that it is viscous, and consequently reacts to the friction of the particles against each other (because of their lateral difference in speed). How does the incompressibility of the continuous medium express itself on its speed field? Thanks to the Reynolds formula, for example. In particular if $f = 1$ in this formula, we note that the volume of $\omega(t)$ varies according to the divergence of u. A volume conserving field has therefore zero divergence.

If the density of external force is f, by writing that the sum of the forces applied on an elementary volume is zero we have

$$\int_{\partial_V} \Sigma n d\sigma + \int_V f dx = 0$$

which using the divergence theorem gives

$$\int_V \operatorname{div} \Sigma dx = -\int_V f dx.$$

Taking into account $\operatorname{div} u = 0$ we compute $\operatorname{div} \Sigma = -\mu \Delta u + \nabla p$. By considering boundary conditions of Dirichlet type on a domain Ω, we find Stokes' equations:

$$\begin{cases} -\mu \Delta u + \nabla p = f & \text{in } \Omega \\ \operatorname{div} u = 0 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

5.2 A first variational formulation

Let $V = \{u \in H_0^1(\Omega)^d,\ \operatorname{div} u = 0\}$. We pick $v \in V$ and multiply the first equation by v and integrate on Ω. By applying Green's formula and the boundary conditions on u and v we obtain

$$\int_\Omega \mu \nabla u : \nabla v dx - \int_\Omega p \operatorname{div} v dx = \int_\Omega f v dx$$

where $\nabla u : \nabla v = \sum_{i=1}^d \nabla u_i \cdot \nabla v_i$. This gives, given the incompressible condition, the variational formulation: Find $u \in V$, such that

$$\forall v \in V, \quad \int_\Omega \mu \nabla u : \nabla v dx = \int_\Omega f v dx.$$

We note that there is no more pressure in this formulation! The existence by the Lax-Milgram theorem is trivial because the bilinear form is the scalar product on H_0^1 and V is a closed subspace of H_0^1. On the other hand, the interpretation of the problem is difficult because of the absence of pressure, and the fact that $\mathcal{D}(\Omega) \not\subset V$. Suppose we have a solution H^2 of the variational formulation and go back using Green's formula to obtain

$$\forall v \in V, \quad \int_\Omega (-\mu \Delta u + f) v dx = 0$$

The trouble is that, unlike the cases previously discussed during the study of boundary problems and the way back from the weak solution to the classical solution, V is not dense in L^2 and the orthogonal of V in L^2 is not reduced to $\{0\}$. For example, any gradient of a function $q \in H^1$ is in $V^\perp$ since from the divergence formula

$$\int_\Omega \nabla q \cdot v dx = -\int_\Omega q \operatorname{div} v dx = 0$$

due to the zero divergence constraint. De Rham's theorem, very deep in analysis, fully characterizes this orthogonal completely:

Theorem 5.1 (de Rham). Let Ω be a smooth open bounded set of class $\mathcal{C}^1$ of $\mathbb{R}^d$; let L be a continuous linear form on H_0^1. Then L vanishes on V if and only if there is $p \in L^2(\Omega)$ such that

$$L(v) = \int_\Omega p \operatorname{div} v dx, \qquad \forall v \in H_0^1(\Omega)^d$$

and p is unique up to a constant.

The difficulty lies in the fact that we cannot directly discretize the space V because it is difficult to build basic functions with zero divergence (however there has had some progress on this side with the introduction of wavelet basis). Consequently we are not going to enforce $\operatorname{div} u = 0$ in the space but to solve the additional equation in weak form.

5.3 A mixed variational formulation

It corresponds to keep u and p and to define for each variable an appropriate space. We then have two unknowns u and p and we choose two test functions (among which one is vectorial) v and q. The variational formulation writes: To find $(u,p) \in (H_0^1(\Omega))^d \times L^2(\Omega)/\mathbb{R}$ such that

$$\begin{cases} \int_\Omega \mu \nabla u : \nabla v dx - \int_\Omega p \operatorname{div} v dx = \int_\Omega f v dx & \forall v \in (H_0^1(\Omega))^d \\ \int_\Omega q \operatorname{div} u dx = 0 & \forall q \in L^2(\Omega)/\mathbb{R} \end{cases}$$

In the above formulation, the quotient space which is $L^2(\Omega)$ functions where we identify two functions whose difference is constant, could be implemented as

$$L_0^2(\Omega) = \left\{ q \in L^2(\Omega), \ \int_\Omega q dx = 0 \right\}.$$

The existence of solution is dealt with thanks to a generalization of Lax-Milgram theorem (see [7])

Theorem 5.2 (Banach-Nečas-Babuška)**.** Let W be a Banach space and let V be a reflexive Banach space. Let a a bilinear continuous form on $W \times V$ and ℓ a linear continuous form on V. Then, the problem:

$$\text{Find } u \in W \text{ such that}, \quad \forall v \in V, \quad a(u,v) = \ell(v)$$

admits one unique solution *if and only if*

$$\exists \alpha > 0, \qquad \inf_{w \in W} \sup_{v \in V} \frac{a(w,v)}{\|w\|_W \|v\|_V} \geq \alpha,$$

$$\forall v \in V, \quad (\forall w \in W, \ a(w,v) = 0) \Longrightarrow (v = 0).$$

Remark 5.1. One particular case of this theorem is when W and V are Hilbert spaces, we will restrict ourselves to this case in this lecture. If moreover $W = V$, then it is easy to check that the coercivity of a implies the two above conditions, which means that the BNB theorem is stronger than Lax-Milgram.

To apply this theory to Stokes problem, we set $V = (H_0^1)^d \times L_0^2(\Omega)$, endowed with the scalar product

$$\langle X, Y \rangle = \langle p, q \rangle_{L^2(\Omega)} + \langle u, v \rangle_{H^1(\Omega)^d}$$

where $X = (u,p)$ and $Y = (v,q)$ are element of V. The bilinear form a is defined on $V \times V$ by

$$a(X,Y) = \int_\Omega \mu \nabla u : \nabla v dx - \int_\Omega p \operatorname{div} v dx + \int_\Omega q \operatorname{div} u dx$$

and the linear form ℓ is defined on V by:

$$\ell(Y) = \int_\Omega f \cdot v dx$$

You can check easily than a and ℓ are continuous on $V \times V$ and V respectively. However, if we would like to apply Lax-Milgram theorem, the coercivity assumption to check would be

$$a(X, X) = \mu \|\nabla u\|^2_{L^2(\Omega)} \geq \alpha \|X\|^2_V$$

which is impossible since there is no p in $a(X, X)$. Thus we have to rely on the stronger form of Lax-Milgram theorem. However, the Stokes problem has a special structure that we will exploit. Indeed, setting $W = (H^1_0(\Omega))^d$ and $M = L^2_0(\Omega)$ we define on $W \times W$:

$$a_0(u, v) = \int_\Omega \mu \nabla u : \nabla v dx$$

and on $W \times M$ the bilinear form

$$b(v, p) = -\int_\Omega p \operatorname{div} v dx.$$

At last, we keep the linear form ℓ on W defined by

$$\ell(Y) = \int_\Omega f \cdot v dx$$

Then the variational formulation of Stokes problem could be written in the specific form: To find $(u, p) \in W \times M$ such that

$$(BF) \quad \begin{cases} a_0(u, v) + b(v, p) = \ell(v) & \forall v \in W \\ b(u, q) = 0 & \forall q \in M \end{cases}$$

This kind of variational formulation where p does not appear in the second equation and where the form b appears in both equations if often called the Brezzi form. For that special form that appears quite often in applications, we have a special form of the BNB theorem:

Theorem 5.3 (Brezzi theorem)**.** Let W and M be two Hilbert spaces. Assume that a_0 is continuous on $W \times W$, b continuous on $W \times M$ and ℓ continuous on W. Assume further that a_0 is coercive on W. Then (BF) has a unique solution *if and only if*:

$$\exists \beta > 0, \forall p \in M, \quad \sup_{v \in W} \frac{b(v, p)}{\|v\|_W} \geq \beta \|p\|_M$$

Proof. We have to check that the (BNB) assumption are equivalent for the bilinear form $a(X, Y) = a_0(u, v) + b(v, p) - b(u, q)$, still with $X = (u, p)$ and $Y = (v, q)$. From the first (BNB) condition we have

$$\forall X \in V, \sup_{Y \in V} \frac{a(X, Y)}{\|Y\|_V} \geq \alpha \|X\|_V$$

which for $X = (0, p)$, since then $a(X, Y) = b(v, p)$, and $\|X\|_V = \|p\|_M$ gives

$$\forall p \in M, \sup_{Y \in V} \frac{b(v, p)}{\|Y\|_V} \geq \alpha \|p\|_M$$

but in $Y = (v, q)$, q is arbitrary thus *a fortiori* for $q = 0$ we obtain

$$\forall p \in M,\ \sup_{v \in V} \frac{b(v,p)}{\|v\|_W} \geq \alpha \|p\|_M$$

which is the assumption made in our theorem. Conversely, assume this assumption on b is fulfilled, we have to prove the two assumptions of BNB. This is admitted in this lecture and we refer to [7]. □

To apply this to the Stokes problem, we need the following Lemma that we admit:

Lemma 5.1. Let Ω be an open, bounded and connected subset of $\mathbb{R}^d$. Then the mapping

$$\operatorname{div} : H_0^1(\Omega)^d \to L_0^2(\Omega)$$

is onto (surjective).

We could justify somehow this result by the following naive argument: Let $p \in L_0^2(\Omega)$ and let seek $v_p \in H_0^1(\Omega)$ such that $\operatorname{div} v_p = p$ under the form $v_p = \nabla\phi$ with $\phi \in H^2(\Omega)$ verifying $\partial_n \phi = 0$ on $\partial\Omega$. Then ϕ is solution of an homogeneous Neumann problem, that we prove existence (and uniqueness up to a constant). Moreover, we have the estimate

$$\exists C > 0, \forall p \in L_0^2(\Omega),\ \|v_p\|_{H^1(\Omega)} \leq C\|p\|_{L^2(\Omega)}$$

Corollary 5.1. Let Ω be an open, bounded and connected subset of $\mathbb{R}^d$. The condition of Brezzi theorem is satisfied for the Stokes problem, which therefore has an unique solution.

Proof. This corresponds to the case $W = (H_0^1(\Omega))^d$, $M = L_0^2(\Omega)$, and $b(v,p) = -\int_\Omega p \operatorname{div} v dx$. Thus we compute

$$\sup_{v \in W} \frac{b(v,p)}{\|v\|_W} = \sup_{v \in W} \frac{\int_\Omega p \operatorname{div} v dx}{\|v\|_W} \geq \frac{\int_\Omega p \operatorname{div} v_p dx}{\|v\|_W} \geq \frac{\int_\Omega p^2 dx}{\|v\|_W} = \frac{1}{C}\|p\|_{L^2(\Omega)}$$

□

Remark 5.2. The two major properties to get existence and uniqueness for the Stokes problem are therefore the **coerciveness** of a_0, which can be transferred to the discrete setting, and the **surjectivity** of div. The latter has no reason to be easy to transfer to the discrete setting, and by the way, we will see that this will bring some constraints on the choice of the finite element degree of u and p.

5.4 Mixed finite element approximation

We follow the presentation of [2]. We set

$$W^h = \left\{ v \in \mathcal{C}(\overline{\Omega})^d, \quad v_{|T_\ell} \in \mathbb{P}_k^d, \quad v = 0 \text{ on } \partial\Omega \right\}, \quad M^h = \left\{ q \in \mathcal{C}(\overline{\Omega})/\mathbb{R}, \quad q_{|T_\ell} \in \mathbb{P}_{k'}^d \right\},$$

so that $W^h \times M^h$ is a subspace of $W \times M$. The variational approximation thus reads

$$\begin{cases} \int_\Omega \mu \nabla u_h : \nabla v_h dx - \int_\Omega p_h \operatorname{div} v_h dx = \int_\Omega f v_h dx & \forall v_h \in W_h, \\ \int_\Omega q_h \operatorname{div} u_h dx = 0 & \forall q_h \in M_h. \end{cases}$$

Let $(\phi_j)_{1\leq j\leq N_1}$ be a basis of W^h and $(\psi_j)_{1\leq j\leq N_2}$ one of M_h, we look for u_h and p_h in the form $u_h = \sum_{j=1}^{N_1} u_h^j \phi_j(x)$ and $q_h = \sum_{j=1}^{N_2} p_h^j \psi_j(x)$. Thus by denoting by $U_h = (u_h^j)$ and $P_h = (p_h^j)$ the bloc system to solve is:

$$\begin{pmatrix} A_h & B_h^T \\ B_h & O \end{pmatrix} \begin{pmatrix} U_h \\ P_h \end{pmatrix} = \begin{pmatrix} F_h \\ O \end{pmatrix}$$

with $A_h^{ij} = \int_\Omega \mu \nabla \phi_i : \nabla \phi_j dx$, $B_h^{ij} = -\int_\Omega \psi_i \operatorname{div} \phi_j dx$ and $F_h^i = \int_\Omega f \phi_i dx$. The matrix of the above system is symmetrical but not definite positive because the vector with 0 on the N_1 first components and 1 on the next N_2 is in the kernel of the associated linear application. However we can show that the system always admits a solution, and that this one is unique with regard to U_h and unique up to a constant for p_h . Indeed the linear system writes

$$\begin{cases} A_h U_h + B_h^T P_h = F_h \\ B_h U_h = 0 \end{cases} \Leftrightarrow \begin{cases} A_h U_h - F_h = -B_h^T P_h \\ U_h \in \ker B_h \end{cases} \Leftrightarrow \begin{cases} A_h U_h - F_h \in \operatorname{im} B_h^T \\ U_h \in \ker B_h \end{cases}$$

We are in finite dimension, so $\operatorname{im} B_h^T = (\ker B_h)^\perp$ and as this last space is the set of orthogonal vectors to $\ker B_h$, finally the linear system is equivalent to

$$\begin{cases} A_h U_h \cdot Z_h = F_h \cdot Z_h, \quad \forall Z_h \in \ker B_h \\ U_h \in \ker B_h \end{cases}$$

It is a variational problem of finite dimension set on $\ker B_h$. It is easy to see that the assumptions of the Lax-Milgram theorem are fulfilled and that the system therefore admits a unique solution U_h. Then P_h is determined up to an element of $\ker B_h^T$.

But beware, it's not that simple: if we choose the degrees k and k' inappropriately, it may very well happen that $\ker B_h^T$ is not not limited to constant vectors, which it necessarily contains:

Proposition 5.1. The kernel $\ker B_h^T$ contains all constant vectors (i.e. with equal components).

Proof. We just have to prove that $R_h \in M_h$ such that R_h has all its components to 1 belongs to $\ker B_h^T$. The associated function $r_h \in M$ will be identically 1. Thus we take $Z_h \in W_h$ and compute $Z_h \cdot B_h^T R_h = B_h Zh \cdot R_h = \int_\Omega r_h \operatorname{div} z_h dx = \int_\Omega \operatorname{div} z_h dx = 0$ by the divergence theorem and homogeneous boundary conditions on z_h. □

Proposition 5.2. When $k = 1$ and $k' = 1$, that is $\mathbb{P}_1$ for velocity and pressure, $\dim \ker B_h^T > 1$ in general. When $k = 2$ and $k' = 1$, that is to say $\mathbb{P}_2$ for the velocity and $\mathbb{P}_1$ on the pressure, $\ker B_h^T$ is of dimension 1 (Taylor-Hood finite element).

Proof. 1. The case $k = k' = 1$. $R_h \in \ker B_h^T$ if we have as above for all $Z_h \in W_h$, $0 = Z_h \cdot B_h^T R_h = B_h Zh \cdot R_h = \int_\Omega r_h \operatorname{div} z_h dx$. Since z_h is $\mathbb{P}_1$, its divergence is constant on each element of the given triangulation (K_i). Thus

$$\int_\Omega r_h \operatorname{div} z_h dx = \sum_i (\operatorname{div} z_h)_{|K_i} \int_{K_i} r_h dx$$

Thus to find a non constant element of $\ker B_h^T$ amounts to cancel each integral on triangle. But as r_h is also $\mathbb{P}_1$, its integral on an triangle is proportional to the mean of its values on vertices. Therefore is suffices to find a function r_h such that the mean value on vertices of triangle is zero. A simple example is below:

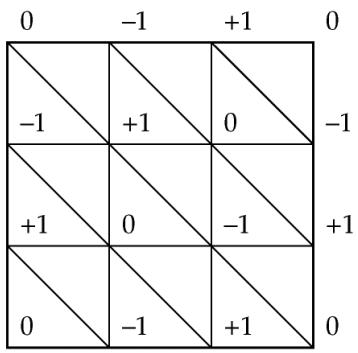


Figure 5.1: Unstable mode of pressure for $\mathbb{P}_1 - \mathbb{P}_1$ finite element on Stokes equations (from [2]).

2. The case $k = 2$, $k' = 1$. Let $R_h \in \ker B_h^T$, we already compute that $\int_\Omega r_h \operatorname{div} z_h dx = 0$ for all $z_h \in W_h$. Integrating by parts gives

$$\forall z_h \in W_h, \quad \int_\Omega \nabla r_h \cdot z_h dx = 0.$$

Since r_h is $\mathbb{P}_1$, its gradient is constant over triangles. Thus

$$\int_\Omega \nabla r_h \cdot z_h dx = \sum_i (\nabla r_h)_{|K_i} \cdot \int_{K_i} z_h dx$$

Now z_h is $\mathbb{P}_2$, we have to rely on a quadrature formula exact on polynomials of degree less or equal to 2. This is the case of the mid-edge formula which states (in dimension 2, but the same formula holds in dimension N with $N(N+1)/2$ instead of 3) that

$$\int_T f(x)dx \approx \frac{|T|}{3}\left[f(a_{12}) + f(a_{13}) + f(a_{23})\right],$$

where $T = (a_1, a_2, a_3)$ is a triangle and a_{ij} the middle of edge $[a_i, a_j]$. Therefore denoting by m_{ij} the middle of edge shared by K_i and K_j, and since z_h cancels on the boundary, we have

$$\sum_i (\nabla r_h)_{|K_i} \cdot \int_{K_i} z_h dx = \frac{1}{3}\sum_i z_h(m_{ij}) \cdot \left[|K_i|(\nabla r_h)_{|K_i} + |K_j|(\nabla r_h)_{|K_j}\right]$$

And for r_h to belong to $\ker B_h^T$, this quantity should vanish for any z_h. If we take z_h equal to e_k on a_{ij} and 0 elsewhere, with (e_k) the canonical basis of R^N, we get that

$$K_i|(\nabla r_h)_{|K_i} + |K_j|(\nabla r_h)_{|K_j}$$

for all i, j. The function r_h should have a gradient constant inside an element but with direction changing direction from an element to another. Since the tangential gradient is continuous in $\mathbb{P}_1$ finite element, this entails that the gradient of r_h is zero everywhere. Thus the kernel of B_h^T contains only constants.

□

5.5 Lid driven cavity flow

As an illustration of this instability occurring when inf-sup condition is not satisfied, we consider the simple following example of driven cavity flow. In this classical test case, we study a flow into a square cavity $\Omega = (0,1)^2$ subject to the following boundary conditions: homogeneous Dirichlet (i.e. zero velocity) on three sides and prescribed tangential velocity on the last one (say for instance, the top). We choose $u(x,y) = (x^2, 0)$ on $y = 1$. The flow created is a vortex inside the box. The boundary problem solved is thus:

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{in } \Omega \\ \operatorname{div} u = 0 & \text{in } \Omega \\ u = x^2 & \text{on } \partial\Omega \cap \{y = 1\} \\ u = 0 & \text{on } \partial\Omega \setminus \{y = 1\} \end{cases}$$

and the approximate variational formulation is given by:

$$\int_\Omega \nabla u_h : \nabla v_h dx - \int_\Omega p_h \operatorname{div} v_h dx - \int_\Omega q_h \operatorname{div} u_h dx = 0, \quad \forall (v_h, q_h) \in W_h \times M_h.$$

To remove the non-uniqueness problem on p, we replace the equation $\operatorname{div} u = 0$ by a regularised version: $\operatorname{div} u + \varepsilon p = 0$ for ε small, which leads to the regularised variational formulation:

$$\int_\Omega \nabla u_h : \nabla v_h dx - \int_\Omega p_h \operatorname{div} v_h dx - \int_\Omega q_h \operatorname{div} u_h - \varepsilon p_h q_h dx = 0, \quad \forall (v_h, q_h) \in W_h \times M_h.$$

Here is the FreeFEM++ code [10] for this simple problem, where we took $\varepsilon = 10^{-6}$:

```
// domain : a square discretized by triangle, 8 nodes by side
mesh Th=square(8,8);
// definition of the velocity component space
fespace Wh(Th,P2);
// definition of the pressure space
fespace Mh(Th,P1);
Wh u2,v2;
Wh u1,v1;
Mh p,q;

// Solve the Stokes problem
solve Stokes (u1,u2,p,v1,v2,q) = int2d(Th)(
        (dx(u1)*dx(v1)+dy(u1)*dy(v1)+dx(u2)*dx(v2)+dy(u2)*dy(v2))
```

```
14          - (0.000001)*p*q-p*dx(v1)-p*dy(v2)-dx(u1)*q-dy(u2)*q )
15        + on(3,u1=x*x,u2=0)
16        + on(1,2,4,u1=0,u2=0);
17
18 // Plot the solution
19 plot(coef=0.5,cmm="Velocity",Th,[u1,u2]);
20 plot(cmm="Pressure", p, fill=true,nbiso=20, value=true);
```

As you can see on lines 4 and 6, the above code corresponds to the case $k = 2$ and $k' = 1$, which satisfies the the kernel condition seen in the former proposition. Therefore we expect a good behaviour. We plot on figure 5.2 the simple mesh, the velocity and pressure field. If we

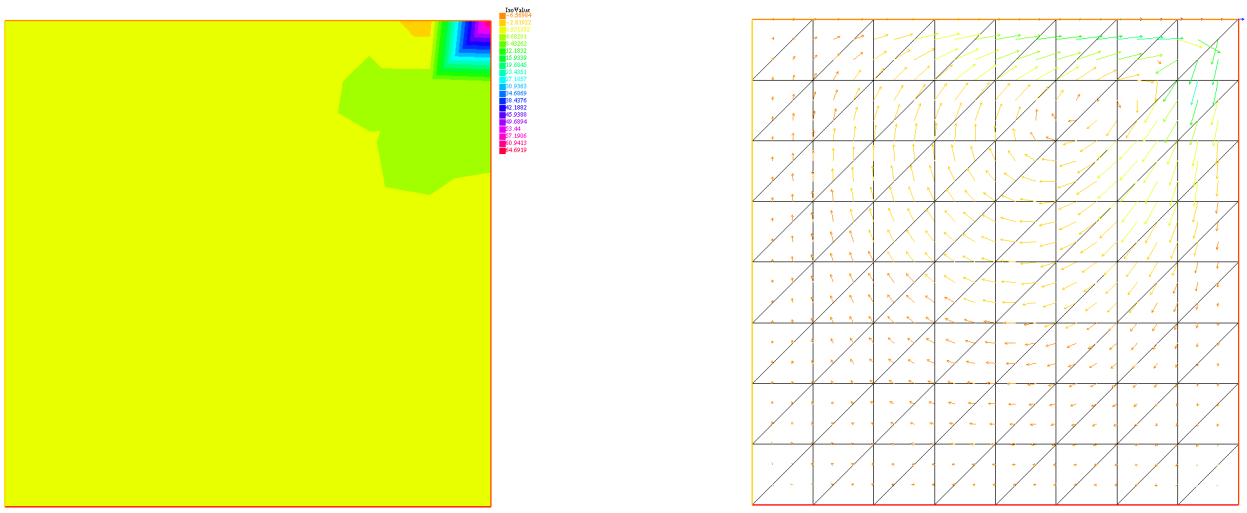


Figure 5.2: Pressure and velocity field (on top of mesh) for $\mathbb{P}_2/\mathbb{P}_1$ approximation of Stokes equations for the driven cavity flow.

consider rather the problematic case $k = 1$ and $k' = 1$ (by changing the 2 by 1 in line 4, we get the checkerboard oscillation in pressure that was predicted in figure 5.1:

Remark 5.3. The $\mathbb{P}_2/\mathbb{P}_1$ or Taylor-Hood approximation for Stokes equation is accurate an verifies the inf-sup condition, but it is also expensive since it involves, in dimension 2 on one element, 18 degrees of freedom (6 for each component of the velocity, and 3 for the pressure). In dimension 3, it reaches 34 degrees of freedom per element, which is quite a lot. A popular element is therefore the $\mathbb{P}_{1b}/\mathbb{P}_1$ where b stands for "bubble". It amounts to add one basis function per element K_i, which is the product of the $\mathbb{P}_1$ basis functions $27\lambda_1^i\lambda_2^i\lambda_3^i$. Since this function is zero on the edges of K_i and positive inside, it looks like a bubble.

If we come back to the formula obtained for $\mathbb{P}_2/\mathbb{P}_1$ approximation:

$$\int_\Omega \nabla r_h \cdot z_h dx = \sum_i (\nabla r_h)_{|K_i} \cdot \int_{K_i} z_h dx$$

we see that this "bubble" function has support strictly included in a given triangle. So since the above formula should hold for any z_h, let us choose $z_h = 27\lambda_1^i\lambda_2^i\lambda_3^i e_k$ where e_k is the k-th basis vector of R^2. Then the equality is $(\partial_{x_k} r_h)_{|K_i} = 0$, which means that r_h is indeed constant in space. We plot below on figure 5.4 the solution for this element.

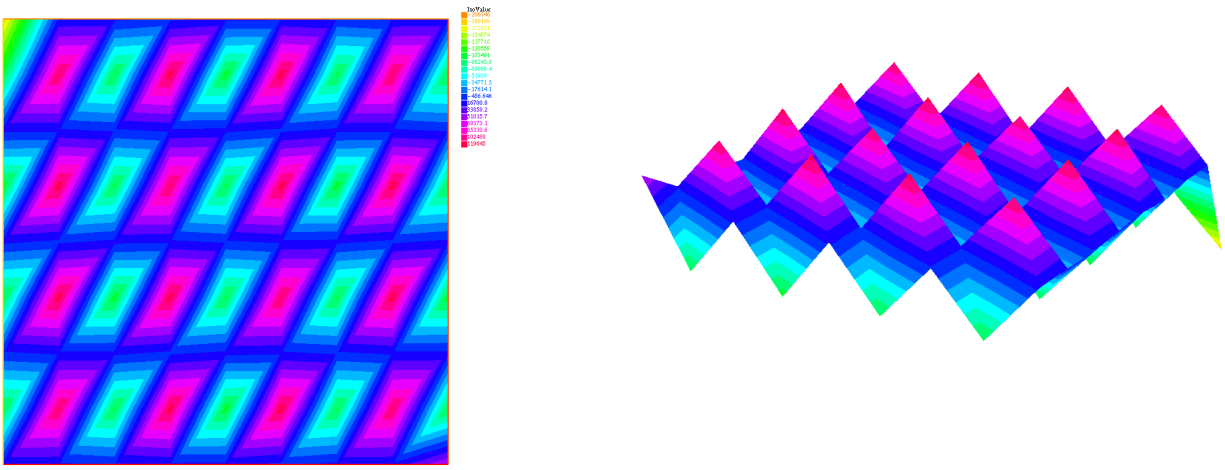


Figure 5.3: Checkerboard instability on pressure for $\mathbb{P}_1/\mathbb{P}_1$ approximation of Stokes equations for the driven cavity flow.

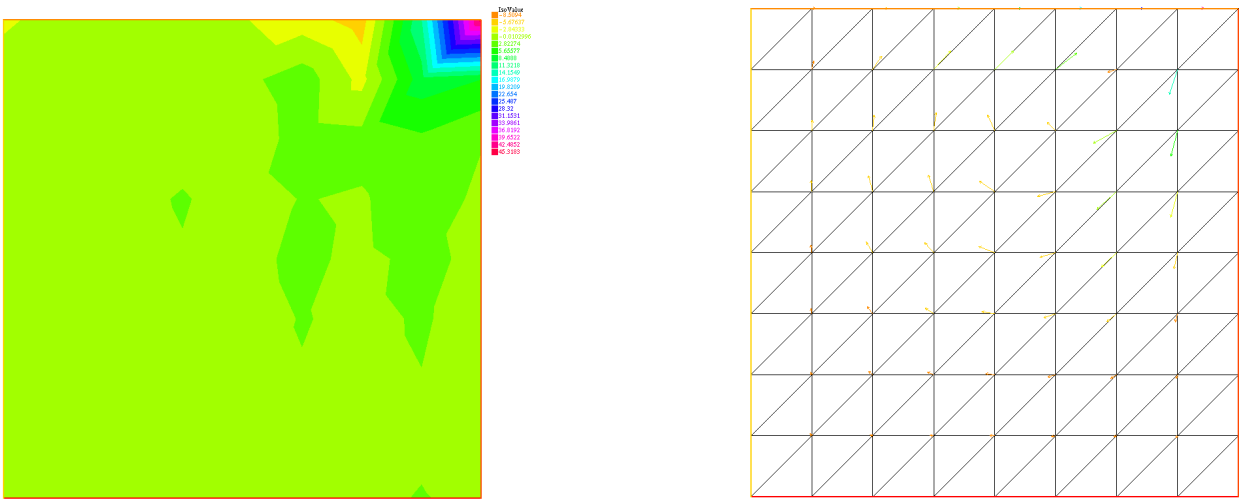


Figure 5.4: Pressure and velocity field (on top of mesh) for $\mathbb{P}_{1b}/\mathbb{P}_1$ approximation of Stokes equations for the driven cavity flow.

Chapter 6

Time-dependent problems

Practically speaking, the time variable is numerically more often dealt with using finite difference schemes. Indeed, a space-time finite element method could be think of, but this would mean meshing a higher dimensional space, which is always costly, and moreover, doing so will produce a method where the unknowns are approximated at different times (or this is essentially a finite difference method) which is hard to cope with. We will consider below several standard time-dependent problems and we refer to practicals for more details.

6.1 Heat equation

Let Ω be an open bounded subset of $\mathbb{R}^d$ and $T > 0$. For $f \in L^2(\Omega \times (0,T))$ and $u_0 \in L^2(\Omega)$ we consider the following problem: To find $u : \Omega \times (0,T) \to \mathbb{R}$ such that

$$(H) \qquad \begin{cases} \partial_t u - \Delta u = f & \text{in } \Omega \times (0,T), \\ u = 0 & \text{on } \partial\Omega \times (0,T), \\ u = u_0 & \text{in } \Omega \times \{0\}. \end{cases}$$

A finite difference in time scheme amounts to introduce a discretization of $(0,T)$ by setting $t_n = n\delta t$, for $n = 0, \dots, N$, with $\delta t = \frac{T}{N}$ and some $N > 0$. Then we can consider several time stepping schemes as for standard differential equations of full finite difference methods. For instance the implicit (also called forward) Euler scheme writes: For $n \geq 0$, find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\frac{u^{n+1} - u^n}{\delta t} - \Delta u^{n+1} = f(t_{n+1}) \quad \text{in } \Omega$$

which in variational form is a standard Dirichlet problem: For $n \geq 0$, find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\forall v \in H_0^1(\Omega), \quad \int_\Omega \delta t \nabla u^{n+1} \cdot \nabla v + u^{n+1} v dx = \int_\Omega (u^n + \delta t f(t_{n+1})) v dx.$$

Likewise, the popular unconditionally stable and second order Crank-Nicolson scheme is given by: For $n \geq 0$, find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\frac{u^{n+1} - u^n}{\delta t} - \frac{1}{2}\Delta u^{n+1} - \frac{1}{2}\Delta u^n = f(t_{n+\frac{1}{2}}) \quad \text{in } \Omega$$

which in variational form is still a basic Dirichlet problem: For $n \geq 0$, find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\forall v \in H_0^1(\Omega), \quad \int_\Omega \frac{1}{2}\delta t \nabla u^{n+1} \cdot \nabla v + u^{n+1} v dx = \int_\Omega (u^n + \frac{1}{2}\delta t f(t_{n+\frac{1}{2}}))v - \frac{1}{2}\delta t \nabla u^n \cdot \nabla v dx.$$

6.2 Wave equation

The situation for the wave equation is quite similar, except that the time differencing is order two. Let Ω be an open bounded subset of $\mathbb{R}^d$ and $T > 0$. For $f \in L^2(\Omega \times (0,T))$ and $u_0, v_0 \in L^2(\Omega)$ we consider the following problem: To find $u : \Omega \times (0,T) \to \mathbb{R}$ such that

$$(W) \qquad \begin{cases} \partial_{tt} u - \Delta u = f & \text{in } \Omega \times (0,T), \\ u = 0 & \text{on } \partial\Omega \times (0,T), \\ u = u_0, \ \partial_t u = v_0 & \text{in } \Omega \times \{0\}. \end{cases}$$

The first schema is explicit and often called the leapfrog scheme for the wave equation. With the same notation as above, for $n \geq 1$, we want to find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\frac{u^{n+1} - 2u^n + u^{n-1}}{\delta t^2} - \Delta u^n = f(t_n) \quad \text{in } \Omega$$

which has to be initialised to compute u^1. Usually to keep second order in time, rather than setting $u^1 = u^0 + \delta t v_0$ which is order one, we introduce u^{-1} by the second order formula $u^1 = u^{-1} + 2\delta t v_0$, and we plug its expression in:

$$\frac{u^1 - 2u_0 + u^{-1}}{\delta t^2} - \Delta u_0 = f(t_0) \quad \text{in } \Omega$$

which gives

$$u^1 = u^0 + \delta t v_0 + \frac{1}{2}\delta t^2 (\Delta u_0 + f(t_0)).$$

This scheme is easy to implement since it is explicit, but has a CFL restriction on time step which is imposed by the smallest element of the triangulation. Another idea would be to replace Δu^n by $\frac{1}{2}(\Delta u^{n+1} + \Delta u^{n-1})$, which would give an implicit and unconditionally stable scheme. Sadly, this scheme suffers from a large numerical dispersion, so that the waves would have poorly approximated velocities.

One widely used implicit scheme for wave equation is the Newmark scheme, which replaces Δu^n in the explicit scheme by $\frac{1}{4}\Delta(u^{n+1} + 2u^n + u^{n-1})$. Then for $n \geq 1$, we want to find $u^{n+1} \in H_0^1(\Omega)$ such that

$$\frac{u^{n+1} - 2u^n + u^{n-1}}{\delta t^2} - \frac{1}{4}\Delta(u^{n+1} + 2u^n + u^{n-1}) = f(t_n) \quad \text{in } \Omega$$

Still, we have to compute the u^1 term by a second order scheme to preserve accuracy. Here we obtain:

$$\frac{u^1 - 2u_0 + u^{-1}}{\delta t^2} - \frac{1}{4}\Delta(u^1 + 2u_0 + u^{-1}) = f(t_0) \quad \text{in } \Omega$$

which gives an elliptic equation to solve to get u^1,

$$u^1 - \frac{\delta t^2}{4}\Delta u^1 = u^0 + \delta t v_0 + \frac{1}{4}\delta t^2(\Delta u_0 - \delta t \Delta v_0 + 2f(t_0)).$$

So the Newmark algorithm, in variational formulation finds first $u^1 \in H_0^1(\Omega)$, knowing u^0 and v^0, solution of

$$\forall v \in H_0^1(\Omega), \quad \int_\Omega u^1 v + \frac{1}{4}\delta t^2 \nabla u^1 \cdot \nabla v dx = \int_\Omega (u_0 + \delta t v_0 + \frac{\delta^2}{2} f(t_0))v - \frac{1}{4}\delta t^2(\nabla u_0 - \delta t \nabla v_0)\cdot \nabla v dx.$$

And then for $n \geq 1$, to find $u^{n+1} \in H_0^1(\Omega)$ such that $\forall v \in H_0^1(\Omega)$,

$$\int_\Omega u^{n+1} v + \frac{1}{4}\delta t^2 \nabla u^{n+1} \cdot \nabla v dx = \int_\Omega (2u^n - u^{n-1} + \delta t^2 f(t_n))v - \frac{1}{4}\delta t^2(2\nabla u^n + \nabla u^{n-1}) \cdot \nabla v dx.$$

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